

Quaternionic division points on curves in the Siegel threefold

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September 24, 2026

Abstract

We prove the quaternionic-division component of Zilber–Pink for curves in $\mathcal{A}_2$ over $\overline{\mathbb{Q}}$. A Hodge-generic algebraic curve contains only finitely many points whose full geometric rational endomorphism algebra is an indefinite quaternion division algebra over $\mathbb{Q}$. No boundary or reduction hypothesis is required.

Contents

1	Introduction	2
1.1	Earlier results	2
1.2	The height argument	3
1.3	Conventions and organization	4
2	The quadratic variation and its isogeny exceptions	4
2.1	Fixed curves and finite covers	4
2.2	A rank-five operator model	5
2.3	Monodromy and rational isogenies	7
2.4	Only finitely many dominating correspondences	8
3	Interpolation with two arguments	11
3.1	Weighted absolute heights	11
3.2	The interpolation statement	11
3.3	Effective algebra in fixed dimension	13
3.4	An ordinary-point zero estimate	14
3.5	The graph and its special fibre	16
3.6	Interpolation by successive powers of the parameter	16
3.7	Product formula and descent	19
4	Arithmetic coordinates and common local relations	21
4.1	Integral endomorphisms and de Rham coordinates	22
4.2	Horizontal comparison on good-reduction disks	25
4.3	Smallness on the actual center branches	27
4.4	Grouping the equations	28
5	Auxiliary markings and the height bound	30
5.1	A finite-field incidence	31
5.2	Fixing the level and the exceptional primes	32
5.3	Nonidentity on every branch	34
5.4	Application of interpolation and descent	37
6	From heights to finiteness	39
6.1	Flat coordinates at all complex embeddings	39
6.2	Counting distinct curve arguments	41
6.3	Descent to the coarse curve	43

1 Introduction

The Siegel moduli variety $\mathcal{A}_2$ parametrizes principally polarized abelian surfaces and has dimension three. An indefinite quaternion division algebra can occur as the full rational endomorphism algebra of such a surface. The corresponding non-CM points lie on special curves. The Zilber–Pink conjecture predicts that a Hodge-generic algebraic curve meets the union of these special curves in only finitely many points. The quaternion algebra, its discriminant, and the endomorphism order are allowed to vary in this prediction.

For a geometric point s of $\mathcal{A}_2$, write (A_s, λ_s) for the represented polarized surface and put

$$\mathrm{End}^0(A_s) = \mathrm{End}_{\overline{\mathbb{Q}}}(A_s) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

A curve is *Hodge generic* if it is not contained in a proper special subvariety, including a Hecke translate of one.

Theorem 1.1. *Let $C \subset \mathcal{A}_{2, \overline{\mathbb{Q}}}$ be a reduced irreducible closed Hodge-generic algebraic curve. Then*

$$\Sigma_{\mathrm{QM}}(C) = \{s \in C(\overline{\mathbb{Q}}) : \mathrm{End}^0(A_s) \text{ is an indefinite quaternion division algebra over } \mathbb{Q}\}$$

is finite.

Thus [Theorem 1.1](#) resolves affirmatively the quaternionic-division component of the Zilber–Pink conjecture for curves in $\mathcal{A}_2$. Its condition is equality of the full geometric endomorphism algebra. In particular, it excludes the split algebra $M_2(\mathbb{Q})$ and does not replace the endomorphism condition by the existence of a quaternion embedding into a larger algebra. No condition is imposed on the boundary of C , and the points are counted across all quaternionic special curves at once.

Throughout the paper, a *QM point* means a point whose full geometric rational endomorphism algebra is an indefinite quaternion division algebra over $\mathbb{Q}$.

1.1 Earlier results

The general unlikely-intersection principle asks when an intersection with a special subvariety has larger dimension than the ambient dimension count predicts. Zilber formulated this principle in the semiabelian setting [[Zil02](#)]; Pink formulated its Shimura counterpart [[Pin05](#), Conjecture 1.3]. For a Hodge-generic curve in a threefold, intersections with special curves have negative expected dimension. Pink’s nondensity prediction therefore requires finiteness even after all such special curves are allowed to vary. The locus in [Theorem 1.1](#) is the part arising from division quaternionic multiplication.

Daw and Orr proved the conclusion when the closure of the curve meets the zero-dimensional boundary stratum of the Baily–Borel compactification [[DO22](#), Theorem 1.4]. Their reduction for an arbitrary curve assumes a large-Galois-orbit estimate [[DO22](#), Theorem 1.3 and Conjecture 6.2]. Their later work proves finiteness against all special curves under that same boundary hypothesis [[DO25](#), Corollary 1.2]. Papas obtained further finiteness results under degeneration in higher-dimensional Siegel varieties, with additional conditions on endomorphism algebras and their behavior under reduction; in dimension two, his boundary hypothesis still requires the zero-dimensional stratum [[Pap24](#), Theorem 1.4].

Papas’s work with v -adic values of G -functions studies neighborhoods of an interior quaternionic point. It gives a height estimate involving the number of supersingular places at which the varying point is close to the chosen center, and finiteness when that number is bounded [[Pap25b](#), Theorem 1.7 and Corollary 1.9]. The horizontal good-reduction comparisons in [[Pap25a](#), Theorem 3.4] are also useful here. We prove a height estimate independent of the number of supersingular proximity places.

1.2 The height argument

Our height argument belongs to the arithmetic auxiliary-function tradition of Bombieri and André [Bom81, And89]. Interior period matrices and their p -adic realizations enter André’s motivic treatment [And95] and Papas’s work cited above. We prove the interpolation statement needed here directly for fixed algebraic differential systems at an ordinary point. In this terminology, “ordinary” means that the differential system has no pole at the center; it imposes no ordinary-reduction condition on the abelian surface.

To rule out infinitely many targets, pass to a fixed smooth fine-level cover and choose an interior quaternionic point t_0 . The substantive estimate is

$$h(t) \leq c_1[K(t) : K]^{c_2} \quad (t \text{ a division-quaternionic point}), \quad (1.1)$$

where K and the ample logarithmic height h are fixed. The constants may depend on the curve. This estimate is proved in Theorem 5.5; it is not an assumed Galois-orbit bound.

The equations come from a rank-five quadratic variation. In the first cohomology of each fiber, take the trace-zero operators that are self-adjoint for the alternating polarization, with pairing $\langle T, U \rangle = \frac{1}{4} \text{Tr}(TU)$. At a QM point, the actual Rosati-symmetric endomorphisms in this space span a positive definite two-plane E_t . We compare a fixed pair of independent integral endomorphisms at the center with a small integral spanning pair at the target. Write B and P for their de Rham coordinates, and $Y(a)$ for horizontal transport from the target back to the center, where $a = x(t)$ is a local algebraic parameter with $x(t_0) = 0$.

At places where the target is close to the center, these coordinates satisfy a matrix of trace relations

$$\langle B, Y(a)P \rangle = T_v \in M_2(\mathbb{Q}).$$

The four entries pair the two centered vectors with the two transported target vectors. At finite places the relations come from endomorphisms of their common good reduction. Although the rational matrix T_v can vary with the prime, a lift of residue Frobenius fixes it. Comparing the relation with its conjugate therefore eliminates T_v . The two arguments are a and $b = aR$, where b is a conjugate of a and $R = b/a$ is included among the algebraic coordinates. The number of conjugate tuples is polynomial in the field degree, so one common system retains a substantial sum of local smallness. The finitely many exceptional primes and the archimedean places are grouped directly by their trace matrices.

The interpolation theorem in Theorem 3.1 turns this local information into (1.1). Its inputs are common analytic equations, controlled coefficient heights, sufficiently small values of a , and nonidentity on every algebraic branch used in its proof. The auxiliary graph is obtained by adjoining the quotient of a finite Taylor truncation of a common relation by a power of a . On this graph, the proof constructs a low-degree polynomial. The product formula forces that polynomial to vanish at the target, while the construction ensures that it does not vanish on the whole graph. Intersection and projection then reduce the dimension of an algebraic variety containing the target. Repeating this descent proves the height bound. An ordinary-point differential estimate makes the required orders polynomial in the algebraic degrees.

It remains to prove the nonidentity input. Full monodromy handles the single-argument relations and pairs of arguments that vary independently. The exceptions are pairs lying on an isogeny correspondence dominating both fixed curves. There are only finitely many such correspondences, up to integral changes of lift. We choose one auxiliary prime ℓ and a fixed level cover on which each target admits a marking with the following property: modulo ℓ , its plane and the centered plane span a degenerate four-space. A functional identity in the exceptional case would force a Frobenius quasi-isogeny to act by one common sign on both planes. The prescribed incidence rules this out, using the polarized multiplier and the

integral trace form. This argument also covers supersingular reductions and target planes that are themselves degenerate modulo ℓ .

Once (1.1) is known, endomorphism estimates and one-variable Hodge-norm bounds give polynomially bounded integral coordinates for two Hodge vectors at every point of a large Galois orbit. The periods lie in the three-dimensional quadric of isotropic lines in the rank-five space. The period at a QM point is orthogonal to its plane E_t , and the orthogonal complement of each positive two-plane cuts this quadric in a conic. The path form of o-minimal counting would produce a one-parameter semialgebraic family of such planes whose conics contain a nonconstant period arc. Their union has algebraic closure of dimension at most two. Full monodromy, however, makes the period image Zariski dense in the three-dimensional quadric, giving the contradiction.

This last step follows the arithmetic point-counting strategy of Pila and Zannier [PZ08]. The rational-point theorem of Pila and Wilkie [PW06], refined into definable blocks by Pila [Pil11, Theorem 3.6], underlies the projected path-counting result of Habegger and Pila [HP16, Corollary 7.2] used here. The numerical bounds for the Hodge vectors come from the endomorphism estimates of Masser and Wüstholz [MW94] and Schmid's degeneration and norm theorems [Sch73]. In the rank-five variation the final geometric contradiction is a direct dimension calculation followed by the monodromy theorem of André [And92].

1.3 Conventions and organization

All varieties and algebraic points are over $\overline{\mathbb{Q}}$ unless a field is specified. Algebraic heights are absolute logarithmic heights. Local absolute values extend the usual, nonsquared absolute values on $\mathbb{Q}$, and local sums carry the normalized weights. The rational coordinate height used in point counting is multiplicative and will be denoted H_{rat} .

Constants and polynomial exponents may depend on fixed curves, covers, centers, frames, and connections. They never depend on the varying quaternion algebra or target point. A polynomial bound always refers to a fixed number of variables and fixed connection ranks.

Section 2 establishes the geometric and monodromy input. The interpolation theorem is proved in Section 3. Section 4 constructs the arithmetic data and common equations. Section 5 proves their nonidentity and the height estimate. Section 6 finishes the proof and descends to the coarse curve.

2 The quadratic variation and its isogeny exceptions

Quaternionic multiplication will be recorded by two independent Hodge vectors in a rank-five quadratic local system. We first construct this system on a fixed fine-level curve and determine its monodromy. For two curve arguments, failure of independent monodromy forces a polarized isogeny relation. We show that only finitely many such relations can dominate the fixed curves. This finiteness will allow the auxiliary level in Section 5 to be chosen before any target point is considered.

2.1 Fixed curves and finite covers

Lemma 2.1 (Passage to fixed covers). *Let $C \subset \mathcal{A}_2$ be a reduced irreducible closed curve over $\overline{\mathbb{Q}}$. There is a finite surjective map $\pi : S \rightarrow C$ from a smooth geometrically connected curve over a number field, together with a principally polarized abelian scheme $f : \mathcal{A} \rightarrow S$ whose coarse moduli map is π . One may obtain S by normalization and a fixed full symplectic level $N \geq 3$.*

If infinitely many points of C have the endomorphism algebra in Theorem 1.1, then infinitely many points of S have that property. Deleting finitely many points of S , or making a

further fixed finite surjective cover and selecting a dominating connected component, preserves this assertion. Once all fixed data are defined over K , degrees and ample logarithmic heights on the smooth completions satisfy

$$[K(\pi(t)) : K] \leq [K(t) : K] \leq D[K(\pi(t)) : K], \quad c^{-1}h_C(\pi(t)) - c \leq h_S(t) \leq ch_C(\pi(t)) + c, \quad (2.1)$$

with fixed constants D, c . The same statements hold for any further fixed cover. In particular, a degree-dependent polynomial height bound obtained on one selected lift of every target descends to the original targets.

Proof. The full level moduli space is a fine moduli scheme with a universal abelian scheme [Cha85, Chapter I, Theorem 1.4]; for the complex moduli interpretation see also [Mil13, Theorem 4.6 and Corollary 4.7]. The finite change-of-level group acts on it, and its quotient is the coarse moduli space, so the forgetful map is finite. We fix the symplectic pairing component after adjoining the relevant roots of unity. Normalize a component of its pullback dominating C . Normal curves in characteristic zero are smooth; the resulting map is finite and surjective. All the schemes and morphisms in this construction descend to a number field. Fibres of a finite map are finite, so an infinite set downstairs has infinitely many lifts. For a cover with several components, at least one component contains infinitely many lifts; a dominating component of a finite cover of an irreducible curve is itself surjective.

After normalizing the base, the finite maps extend to finite maps between smooth projective completions. For such a map of generic degree D_0 , the degree formula for a fibre gives $[K(t) : K(\pi(t))] \leq D_0$. Over the finitely many singular points of the original coarse curve, the fixed finite map has a fixed finite set of preimages; enlarge D to bound their residue-field extensions as well. Thus the constant D in (2.1) is uniform, but need not equal the generic degree of the map to the singular curve.

Functoriality of heights identifies a height of a pulled-back ample divisor with the height downstairs up to a bounded function. Any two ample divisors on a fixed projective curve bound each other's heights linearly: sufficiently large positive integral multiples of either minus the other are ample and have heights bounded below. This proves (2.1). Replacing the fixed field K changes degrees by at most a fixed factor. None of these constants depends on the point. $\square$

Assume the target locus is infinite, as will be appropriate for the final contradiction, and choose S as in Lemma 2.1. We write $\bar{S}$ for its smooth projective completion. After any required finite deletions, choose a target $t_0 \in S(\bar{\mathbb{Q}})$ and enlarge the fixed number field K so that $t_0 \in S(K)$. A rational function $x \in K(S)$ can be chosen with a simple zero at t_0 . On a neighbourhood of t_0 , it is an étale parameter; its inverse germ through t_0 is always the germ used below. Other zeros of x do not specify this germ. Rational frames and affine coordinates may be made regular on this neighbourhood by further finite deletions away from t_0 . The finitely many conjugates of these fixed data will also be treated as fixed data. No property of the reduction of t_0 is imposed here.

2.2 A rank-five operator model

Let $\mathbb{H}_{\mathbb{Z}} = R^1 f_* \mathbb{Z}$ on $S(\mathbb{C})$, and let ψ be its alternating polarization. The corresponding rational local system is $\mathbb{H} = \mathbb{H}_{\mathbb{Z}} \otimes \mathbb{Q}$. The adjoint of an operator for ψ is denoted by $T^\dagger$. We use the weight-zero local system

$$\mathbb{V} = \{T \in \text{End}(\mathbb{H}) : T^\dagger = T, \text{Tr}(T) = 0\}, \quad \langle T, U \rangle = \frac{1}{4} \text{Tr}(TU), \quad q(T) = \langle T, T \rangle. \quad (2.2)$$

Its natural integral lattice is the intersection with $\text{End}(\mathbb{H}_{\mathbb{Z}})$. The rational form in (2.2) is integral and unimodular on this lattice after inverting 2. Using another fixed commensurable integral lattice only enlarges the finite set of excluded auxiliary primes.

Lemma 2.2 (The quadratic model and the quaternionic plane). *The local system $\mathbb{V}$ has rank five, a nondegenerate form of real signature $(3, 2)$, and Hodge numbers $(1, 3, 1)$ in types $(1, -1), (0, 0), (-1, 1)$. It is the primitive part of $R^2 f_* \mathbb{Q}(1)$. Conjugation of operators gives a surjective morphism of algebraic groups*

$$\rho : \mathrm{GSp}(\mathbb{H}, \psi) \longrightarrow \mathrm{SO}(\mathbb{V}, q), \quad g : T \longmapsto gTg^{-1}, \quad \ker \rho = \mathbb{G}_m. \quad (2.3)$$

Its restriction to Sp_4 is the spin covering, with kernel $\{\pm \mathrm{id}\}$.

At a point t whose full geometric endomorphism algebra is an indefinite quaternion division algebra, the trace-free Rosati-symmetric endomorphisms form a rational two-plane $E_t \subset \mathbb{V}_t$ of Hodge vectors. The restriction of q to E_t is positive definite. This plane is spanned by two actual integral trace-free symmetric endomorphisms. If $\Lambda_t = \mathbb{V}_{\mathbb{Z}, t}$, then $L_t = E_t \cap \Lambda_t$ is saturated in Λ_t ; a selected generating pair need not generate this saturated lattice.

Proof. In a symplectic basis, the matrix of ψ is $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$. The equations $T^t J = J T$ and $\mathrm{Tr}(T) = 0$ give

$$T = \begin{pmatrix} A & \beta J_2 \\ \gamma J_2 & A^t \end{pmatrix}, \quad A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

One checks that $T^2 = (a^2 + bc - \beta\gamma)I_4$. Thus

$$q(T) = a^2 + bc - \beta\gamma,$$

which has signature $(3, 2)$ and is nondegenerate over $\mathbb{Z}[1/2]$. The alternating form $(u, v) \mapsto \psi(Tu, v)$ identifies the self-adjoint operators with alternating two-forms. The identity operator corresponds to ψ , and its orthogonal complement for the trace form is the primitive part. Taking account of the polarization twist gives $\mathbb{V} \simeq (\wedge^2 \mathbb{H})(1)_{\mathrm{prim}}$ and the stated Hodge numbers.

Conjugation by a similitude preserves self-adjointness and the trace pairing. Its determinant on $\mathbb{V}$ is 1, since GSp_4 is connected. The induced Lie algebra map $\mathfrak{sp}_4 \rightarrow \mathfrak{so}_5$ is nonzero. Simplicity and equality of dimensions make it an isomorphism. Consequently the induced map on the adjoint groups is an isomorphism; these are the split adjoint groups of types C_2 and B_2 . Its kernel on GSp_4 is exactly the scalar torus, proving (2.3) and the assertion about the spin covering. These statements are over $\mathbb{Q}$, not merely over $\mathbb{C}$.

Put $D = \mathrm{End}^0(\mathcal{A}_t)$. The polarization induces the positive Rosati involution on D . Over $\mathbb{R}$, the identification $D \otimes \mathbb{R} \simeq M_2(\mathbb{R})$ turns a positive involution into the adjoint for a positive definite form, and hence into transpose in a suitable basis. Its fixed subspace has dimension three. The cohomological trace is nonzero on the identity, so its kernel in that subspace has dimension two. For a nonzero symmetric endomorphism u , Rosati positivity gives $\mathrm{Tr}(uu^\dagger) = \mathrm{Tr}(u^2) > 0$. Thus q is positive definite on this trace-free subspace.

The rational realization of endomorphisms on H^1 is faithful and identifies endomorphisms with rational Hodge endomorphisms [Mil95, Theorem 4.1]. In particular these vectors are Hodge vectors in $\mathbb{V}_t$. Clearing denominators in a rational basis gives integral endomorphisms with the same symmetry and trace conditions. Equivalently one can first apply $u \mapsto u + u^\dagger$ and then $v \mapsto 4v - \mathrm{Tr}(v)\mathrm{id}$ to integral endomorphisms; the principal polarization makes the first operation integral. Finally, intersection of an integral lattice with a rational subspace is saturated. $\square$

Write $\mathcal{V}_{\mathrm{dR}}$ for the algebraic de Rham realization, with its Gauss–Manin connection and trace form. The connection is the one constructed from the relative de Rham complex [KO68, Theorem 1]; its functoriality and compatibility with cup products follow from that

construction. Algebraic–analytic de Rham comparison [Gro66, Theorem 1’], applied to the relative complexes, identifies its horizontal local system with the Betti realization; this relative interpretation is recorded in [Kat70, Section 0.0]. Actual endomorphisms have compatible Betti, de Rham, and étale realizations, and their trace pairings are rational. For a polarized isogeny $\varphi : A \rightarrow A'$ with $\varphi^*\lambda' = m\lambda$, cohomological pullback $\varphi^* : H^1(A') \rightarrow H^1(A)$ has multiplier m and reverses direction. Our covariant transport on H^1 is

$$T_\varphi = m(\varphi^*)^{-1} : H^1(A) \longrightarrow H^1(A'),$$

which again has multiplier m . We put $I_\varphi(u) = T_\varphi u T_\varphi^{-1}$ in the operator model. This is the isometry in (2.3) in the direction from A to A' ; on actual endomorphisms it realizes $\alpha \mapsto \varphi \alpha \varphi^{-1}$. These conventions also apply to polarized quasi-isogenies with rational multiplier and commute with composition. Thus an inverse isogeny of multiplier m composed with Frobenius of multiplier p has multiplier p/m on this covariant weight-one realization. For an isogeny, $\deg \varphi = m^2$.

Under a flat trivialization, the period is the line of type $(1, -1)$ in the smooth projective quadric

$$Q_V = \{[z] \in \mathbb{P}(V_{\mathbb{C}}) : q(z) = 0\}, \quad \dim_{\mathbb{C}} Q_V = 3. \quad (2.4)$$

A rational vector is of type $(0, 0)$ if and only if it is orthogonal to this line: its reality supplies orthogonality to the conjugate line as well. In particular, the period at a quaternionic target is orthogonal to E_t .

2.3 Monodromy and rational isogenies

Proposition 2.3 (Single and product monodromy). *For the family over any fixed smooth finite cover of a Hodge-generic curve in $\mathcal{A}_2$, the connected algebraic monodromy on $\mathbb{H}$ is Sp_4 , and that on $\mathbb{V}$ is $\mathrm{SO}(V, q)$. The same holds after pullback by a dominant morphism from an irreducible complex algebraic variety, on a smooth open of a resolution.*

Let S_1, S_2 be two such curves, and let an irreducible complex algebraic variety W map dominantly to both. On the sum of the two pulled-back weight-one systems, either the connected monodromy is $\mathrm{Sp}_4 \times \mathrm{Sp}_4$, or the image of W in the product of the two moduli curves is contained in a dominating component of a polarized isogeny correspondence. In the latter case the corresponding local period relation is given by a fixed element of $\mathrm{GSp}_4(\mathbb{Q})$ with positive multiplier.

Proof. The generic Mumford–Tate group of a Hodge-generic curve is GSp_4 . The connected monodromy normality theorem for a polarizable geometric variation on a smooth algebraic base places connected monodromy normally in its derived group [And92, §5, Theorem 1]. Since Sp_4 is $\mathbb{Q}$ -simple, the connected monodromy is either trivial or Sp_4 . In the first case a finite cover kills monodromy, and the theorem of the fixed part makes the variation constant; the moduli map would be constant. This is impossible. This consequence is also recorded in [And92, §6, Remark 1]. A dominant pullback has the same generic Mumford–Tate group and still has a nonconstant moduli map, so the argument also applies there. The assertion for $\mathbb{V}$ follows from (2.3).

For the two systems let M be their generic Mumford–Tate group and H their connected monodromy, after fixing rational symplectic bases on a simply connected open. Both projections of H are Sp_4 . Put $G = \mathrm{PGSp}_4$. The algebraic Goursat lemma and simplicity give two possibilities for the image $\bar{H} \subset G \times G$: it is the product, or it is the graph of a $\mathbb{Q}$ -defined automorphism α of G . Indeed the kernels of the two projections are normal; in an adjoint simple group a proper such kernel is trivial. In the product case dimensions give $H = \mathrm{Sp}_4 \times \mathrm{Sp}_4$.

In the graph case, every automorphism of the split adjoint group of type C_2 is inner, since its Dynkin diagram has no automorphism. As G is adjoint, a $\mathbb{Q}$ -defined inner automorphism is conjugation by an element $\bar{g} \in G(\mathbb{Q})$. The central exact sequence

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{GSp}_4 \longrightarrow G \longrightarrow 1$$

and $H^1(\mathbb{Q}, \mathbb{G}_m) = 0$ show that $\bar{g}$ lifts to $g \in \mathrm{GSp}_4(\mathbb{Q})$. Conjugate the second factor by this lift, so that $\bar{H}$ becomes the diagonal. Normality of H in M forces the adjoint image of M into the normalizer of this diagonal. That normalizer is the diagonal itself: a normalizing pair (g_1, g_2) has $g_2^{-1}g_1$ central in the adjoint group.

The two Hodge homomorphisms therefore agree after passage to G . Their discrepancy on the weight-one spaces is a real algebraic character of the Deligne torus $\mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ into the scalar $\mathbb{G}_m$. Such a character is a power of the norm. The two weights are equal, so restriction to real scalars makes this power zero. The rational map g thus identifies the Hodge structures themselves. Its similitude multiplier is positive because it compares the positive definite forms supplied by their polarizations and complex structures.

This argument holds at very general points of the chosen open; holomorphicity extends the resulting fixed period relation across it. Write $\mu > 0$ for the multiplier of $g : H^1(A_1) \rightarrow H^1(A_2)$. The homology map in the direction from A_1 to A_2 is

$$F = \mu(g^{-1})^\vee : H_1(A_1, \mathbb{Q}) \longrightarrow H_1(A_2, \mathbb{Q}).$$

It has multiplier μ and commutes with the complex structures. Choose an integer $n > 0$ such that nF is integral. It induces a holomorphic isogeny $\varphi : A_1 \rightarrow A_2$, which is algebraic since the tori are projective. Its multiplier is the positive integer $m = n^2\mu$: integrality follows by evaluating its polarization identity on a pair of integral homology vectors of pairing 1. Moreover $\varphi^* = n\mu g^{-1}$, so the covariant cohomological transport defined above is

$$T_\varphi = m(\varphi^*)^{-1} = ng.$$

Thus φ realizes the original projective rational representative.

A fixed rational similitude defines a closed algebraic Hecke correspondence, finite over either moduli factor. Concretely, for a fixed integral multiplier m , its kernels are among the finite subgroup schemes of $A[m]$, and the principal polarizations and fixed level structures give the corresponding finite moduli problem. Closure of the local relation consequently contains the image of W in one such correspondence. Its projections to both curves are dominant by hypothesis. Finite central twists of the local systems can be killed by a finite cover and do not change this conclusion about their moduli points. $\square$

2.4 Only finitely many dominating correspondences

Here the finiteness concerns whole correspondences dominating two fixed curves. Individual isolated isogenous pairs need not belong to the finite list.

Proposition 2.4 (Finitely many isogeny exceptions). *Fix finitely many closed irreducible Hodge-generic curves in $A_2(\mathbb{C})$. For every ordered pair of these curves there are finitely many rational positive projective symplectic similitudes carrying a lift of the first onto a lift of the second, modulo integral changes of lifts. Consequently all dominating polarized isogeny correspondences between this list of curves are contained in a finite union of closed curves in their pairwise products. There is a fixed finite set of positive integers m such that every pair on this union is joined by a polarized isogeny of one of these multipliers, of degree m^2 . Refining the fixed level does not enlarge the required set of underlying multipliers.*

Proof. We first show that the real stabilizer of a lifted curve is discrete. Finite area of the curve quotient will then make its integral stabilizer a subgroup of finite index. Maps between two lifts consequently have only finitely many cosets, and rational representatives will give the fixed isogeny multipliers.

Let $\mathfrak{H}_2$ be Siegel space, let $G_{\mathbb{R}} = \mathrm{PGSp}_4(\mathbb{R})^+$ be its effective connected group of holomorphic isometries, and fix a torsion-free principal congruence subgroup Γ at fine level. For any curve in the list, choose an irreducible analytic component X of its full inverse image in $\mathfrak{H}_2$. It is a closed analytic curve. Its normalization may be used wherever a smooth curve is needed. The integral stabilizer

$$\Gamma_X = \{\gamma \in \Gamma : \gamma X = X\}$$

contains the image of the monodromy of the normalization of the corresponding fine-level curve. In fact this image is the deck group on the lifted normalization over the smooth part. By [Proposition 2.3](#), it is Zariski dense in the real adjoint group.

We verify that the real stabilizer

$$H_X = \{g \in G_{\mathbb{R}} : gX = X\}$$

is discrete. It is closed: if $g_n \rightarrow g$ with $g_n X = X$, closedness of X gives $gX \subset X$, and applying the same argument to g_n^{-1} gives equality. Thus H_X is a Lie subgroup. Its Lie algebra is invariant under Γ_X . Stabilizing a fixed real vector subspace of the adjoint representation is an algebraic condition, so Zariski density makes this Lie algebra an ideal of $\mathfrak{sp}_4(\mathbb{R})$. Simplicity leaves zero or the full Lie algebra. The latter would give $H_X = G_{\mathbb{R}}$, forcing $X = \mathfrak{H}_2$ by transitivity. The dimensions differ, so H_X is discrete.

Discreteness alone does not bound the number of cosets of Γ_X . For this we use that $\Gamma_X \backslash X$, with normalization understood, has finite area for the metric induced from an invariant Hermitian metric on $\mathfrak{H}_2$. To check both the quotient and the effect of singularities, let S_X be the normalization of the corresponding closed fine-level curve and form

$$Y = S_X \times_{\Gamma \backslash \mathfrak{H}_2} \mathfrak{H}_2.$$

The normalization map from S_X to its closed fine-level image is finite, so base change makes $Y \rightarrow \mathfrak{H}_2$ finite, with image the full inverse image of that curve. On the other hand, $Y \rightarrow S_X$ is a covering and locally a biholomorphism, so Y is smooth. The finite map to that inverse image is an isomorphism above the smooth locus of the curve; hence it is its normalization.

The component Y_0 normalizing X has stabilizer Γ_X ; this group acts transitively on every fibre over S_X , since a deck transformation carrying two points of the same fibre to one another preserves their common connected component, and $\Gamma_X \backslash Y_0 \simeq S_X$. It therefore suffices to bound the pullback metric at the finitely many punctures of S_X .

For completeness, any holomorphic map from the unit disk to $\mathfrak{H}_2$ contracts its invariant metric up to a fixed constant relative to the Poincaré metric. Realize $\mathfrak{H}_2$ as a bounded domain. An automorphism moving the image of the origin to a fixed origin, followed by the Cauchy derivative estimate in bounded coordinates, bounds the derivative at zero uniformly. Disk automorphisms give the pointwise assertion. Apply it on the universal cover of a punctured parameter disk. The descended upper bound is a constant multiple of

$$\frac{|du|^2}{|u|^2 \log^2(1/|u|)},$$

whose area on $0 < |u| < \varepsilon$ is finite. On the remaining compact part the period map is holomorphic and has finite area. Zeros of its derivative or singular branches only decrease this upper bound.

We now deduce the precise group-theoretic consequence

$$[H_X : \Gamma_X] < \infty. \quad (2.5)$$

The action of $G_{\mathbb{R}}$ on $\mathfrak{H}_2 = G_{\mathbb{R}}/K_{\max}$ is proper, since $K_{\max}$ is compact. Therefore the discrete group H_X acts properly on X , with finite point stabilizers. Choose a regular point $z \in X$ and a sufficiently small $H_{X,z}$ -invariant disk $D \subset X$ of positive induced area such that $gD \cap D$ is empty for $g \notin H_{X,z}$. Distinct double cosets in $\Gamma_X \backslash H_X / H_{X,z}$ give disjoint images of translates of D in $\Gamma_X \backslash X$. Each translate embeds there: a further identification would give a nonidentity element of the torsion-free group Γ_X conjugate to an element of the finite group $H_{X,z}$. All these disks have the same positive area. Finite total area thus bounds the number of double cosets, and finiteness of $H_{X,z}$ bounds the number of left cosets. This proves (2.5).

Choose lifts X_1, X_2 for two curves. If a rational positive projective similitude g_0 carries X_1 onto X_2 , the set of all real such maps is $g_0 H_{X_1}$. By (2.5) this set has only finitely many classes modulo right multiplication by Γ_{X_1} . Every class containing a rational map admits a rational representative. Integral changes of the chosen lifts add no classes: at coarse level all lifts are conjugate under the full integral symplectic group; at the chosen fine level there are only finitely many resulting classes. Thus there are finitely many rational maps for the whole fixed list.

A dominating component of a Hecke correspondence determines a local equality between a rational translate of one lifted curve and another lifted curve. Both are irreducible closed analytic curves, so this local equality extends to equality of those lifts. It is therefore represented by one of the finite maps just obtained. Lift each projective rational representative to $\mathrm{GSp}_4(\mathbb{Q})$, using $H^1(\mathbb{Q}, \mathbb{G}_m) = 0$, and clear denominators on the integral homology lattices. Its multiplier becomes a positive integer m . This gives an isogeny of multiplier m on the local correspondence. To retain that multiplier after taking closure, work over the fixed fine moduli space in characteristic zero. Subgroups $K \subset A[m]$ that are maximal isotropic for the Weil pairing and have order m^2 form a finite étale scheme over this space: étale-locally the torsion is a constant finite group. Each such subgroup gives the quotient $B = A/K$ with its principal polarization characterized by $q^* \lambda_B = m \lambda_A$. The resulting map to $\mathcal{A}_2 \times \mathcal{A}_2$ is finite. Indeed it is finite over the first coarse factor, hence proper into the product because the second factor is separated, and its fibres are finite. Its image is therefore closed, and every geometric point of that image admits an isogeny of multiplier m . The closure of the local correspondence lies in this image, including at singular coarse points and points with extra automorphisms.

Taking all representatives and the finitely many curve components of their intersections with the pairwise products that dominate both factors gives the required finite union. Additional level structures change only the finite covers of these moduli problems, not the underlying isogenies or their multipliers. $\square$

We apply Proposition 2.4 to the full finite list of absolute Galois conjugates of the original closed coarse curve C . The canonical Siegel moduli space has reflex field $\mathbb{Q}$, and Galois conjugation permutes its special subvarieties [Orr21, Corollary 5.3, arXiv version 2]. If a conjugate of C lay in a proper special subvariety, inverse conjugation would contradict the Hodge genericity of C . Thus every curve in this list is Hodge generic. Enlarging K for centers or markings adds no coarse image to this list. The resulting finite set of multipliers, the chosen center, and the rank-five lattices are fixed before the prime ℓ and the marking cover of Proposition 5.2 are selected. Refining level changes the markings over these same coarse images; the exceptional isogenies are required only between the underlying polarized abelian varieties. Later estimates may therefore be proved on that fixed cover and descended by Lemma 2.1.

3 Interpolation with two arguments

This section proves a polynomial height bound from common analytic relations at a weighted collection of places. The relations involve a small algebraic parameter a , auxiliary algebraic coordinates U , and solutions of fixed ordinary differential systems. A second argument may be aR , where R is a coordinate of U . The conclusion will bound a height parameter h polynomially in a complexity parameter d , provided the relations are not all identities on any of the algebraic branches used in a dimension descent.

3.1 Weighted absolute heights

For a tuple $z = (z_i)$ over a number field F , write $\|z\|_v = \max_i |z_i|_v$ and put

$$h(z) = \sum_v w_v \log \max(1, \|z\|_v), \quad w_v = \frac{[F_v : \mathbb{Q}_v]}{[F : \mathbb{Q}]}.$$

At the infinite places the absolute value is the usual, unsquared one; thus a complex place has weight $2/[F : \mathbb{Q}]$. The same notation for a polynomial, or a finite list of polynomials, means the height of its complete coefficient tuple *with the additional coordinate 1*. It therefore bounds the scale of the coefficients. For each fixed number of variables,

$$\log |G(z)|_v \leq \log \max(1, \|\text{coeff } G\|_v) + \deg(G) \log \max(1, \|z\|_v) + \varepsilon_v \log \# \text{mon}(G), \quad (3.1)$$

where ε_v is 1 at an infinite place and 0 otherwise. In particular, the sum of the last terms is controlled without counting the finite places.

Occasionally the selected local data distinguish embeddings inducing the same place. We then use embeddings $F \hookrightarrow \mathbb{C}$ or $F \hookrightarrow \mathbb{C}_p$, each with weight $1/[F : \mathbb{Q}]$, which gives the same height formula. A selection of embeddings is allowed. On replacing F by a finite extension, we always take *all* extensions of each selected embedding. The sum of their weights is its original weight. All selected sums below are consequently unchanged by such an extension. No bound for the degree of a normal closure or of an auxiliary coefficient field will be used.

3.2 The interpolation statement

In the statement, the finite list of curves, connections, ordinary centres, rational frames regular at the centres, and algebraic chart germs is fixed. Its ranks and all polynomial-combination degrees are fixed. An allowed analytic expression is a polynomial combination of the algebraic coordinates (a, U) , these chart germs, and entries of the horizontal solution matrices, evaluated at a and possibly at aR , where R is one of the coordinates of U . Constant summands are allowed. The total degree in all polynomial factors, including a and every coordinate of U , is bounded by a fixed constant. Only substitution into the fixed germs produces Taylor coefficients of growing degree. The scalar coefficients of the combination may be algebraic and vary with the target.

Theorem 3.1 (Common-relation interpolation). *Fix the analytic data just described, a fixed affine ambient space with coordinates (a, U) , and an algebraic ambient locus X in that space. Let $h, d \geq 2$, let $\Psi(h) = 1 + (\log h)^c$ for a fixed $c \geq 1$, and let $z_* = (a_*, U_*) \in X(\overline{\mathbb{Q}})$ with $a_* \neq 0$. Suppose that the following bounds hold, with fixed polynomials Q_i having nonnegative coefficients.*

- (i) *The ambient locus has equations of degree at most $Q_0(d)$ and affine height at most $Q_0(d)\Psi(h)$, and $h(z_*) \leq Q_0(d)h$.*

(ii) *There is a finite weighted selection $\mathcal{V}$ of places or embeddings for which $|a_*|_v < 1$ and*

$$S_{\text{small}} := \sum_{v \in \mathcal{V}} w_v \log \frac{1}{|a_*|_v} \geq \frac{h}{Q_1(d)}, \quad \sum_{v \in \mathcal{V}} w_v \log \max(1, \|U_*\|_v) \leq Q_2(d)\Psi(h). \quad (3.2)$$

If the second argument occurs, $|R_|_v = 1$ for these local data.*

(iii) *A common list of fixed length s of allowed expressions*

$$E_\nu(a, U) = \sum_{j \geq 0} a^j e_{\nu j}(U), \quad 1 \leq \nu \leq s,$$

vanishes at z_ at every selected local datum. Its coefficients satisfy*

$$\deg e_{\nu j} \leq C(1 + j), \quad h((e_{\nu j})_{\nu, 0 \leq j \leq l}) \leq Q_3(l, d)\Psi(h).$$

There are $b_v \geq 0$ with $\sum_{v \in \mathcal{V}} w_v b_v \leq Q_4(d)\Psi(h)$ such that the convergent tails at the target satisfy, for every $l \geq 0$,

$$\left| \sum_{j \geq l} a_*^j e_{\nu j}(U_*) \right|_v \leq |a_*|_v^l \exp((1 + l)b_v). \quad (3.3)$$

(iv) *If $W \subseteq X$ is an irreducible algebraic variety containing z_* on which a is nonconstant, then on every normalization branch above the generic point of every component of $W \cap V(a)$ the expressions $E_1, \dots, E_s$ are not all identically zero. The expressions use their specified ordinary chart germs.*

Then $h \leq C_1 d^{C_2}$, with constants depending only on the fixed data and the polynomials in the hypotheses.

The hypothesis in (iv) includes branches on proper subvarieties of X . The proof only uses the varieties encountered in the descent below, so verification on those varieties suffices. In the application this is the role of [Proposition 5.4](#).

Here is the descent mechanism. Suppose that h is larger than a polynomial in d , and let $W \subseteq X$ be an irreducible variety containing the target. When a is nonconstant and W meets $a = 0$, we pass to a graph W' that is isomorphic to W over $a \neq 0$. On this graph we construct a polynomial P with

$$P \notin I(W'), \quad P(z'_*) = 0,$$

where z'_* is the lifted target. The intersection with $P = 0$ then projects to a smaller-dimensional variety containing z_* . We discard the graph coordinate after each step, so the ambient dimension stays fixed.

The two requirements on P have different sources. Nonidentity in (iv), together with an ordinary-point zero estimate, gives a graph whose special fibre satisfies a codimension-two condition after one further equation is imposed. Counting coefficients layer by layer in powers of a then gives a polynomial nonzero on W' whose target value is bounded locally by $|a_*|_v^n$ times controlled factors. The ratio of n to its degree L can exceed any prescribed polynomial in d , while both n and L remain polynomially bounded. At the selected places, the factor $|a_*|_v^n$ contributes $-nS_{\text{small}}$ to the logarithm of its value. At the other places, its degree costs at most L times the global height of the lifted target. For a sufficiently large ratio n/L , the product formula forces $P(z'_*) = 0$. The remaining errors are polynomial multiples of $\Psi(h)$ and are absorbed when h is large.

3.3 Effective algebra in fixed dimension

The descent needs two kinds of algebraic bounds: equations for each smaller variety must retain controlled degree and height, and the interpolation polynomial must have a membership identity with controlled coefficients. We record both bounds here. Throughout this subsection, “controlled” means polynomial in the input degrees and linear in $1 +$ the joint logarithmic coefficient height, with ambient dimension fixed.

In a fixed number of variables, ideals with generators of degree at most D have Gröbner bases of degree polynomial in D , for any prescribed monomial order [Dub90, Corollary 8.3]. We may first discard generators linearly dependent on the others, leaving polynomially many. If a polynomial belongs to such an ideal, the degrees in a membership representation are polynomial in D and the degree of that polynomial [Asc04, Theorem 3.4]. This is a bound for arbitrary ideal membership.

These degree bounds also give coefficient bounds from the original input coefficients. Let δ be the Gröbner-basis degree bound, and choose the minimal monomial generators M_α of the initial ideal. Each is the leading monomial of a polynomial of degree at most δ in the ideal. Seek such a polynomial in the form

$$H_\alpha = M_\alpha + \sum_{\substack{N < M_\alpha \\ \deg N \leq \delta}} c_{\alpha N} N, \quad H_\alpha = \sum_i A_{\alpha i} G_i.$$

The membership bound supplies a polynomial bound for the degrees of the $A_{\alpha i}$. The displayed coefficient equations are therefore a solvable affine linear system of polynomial size whose entries are only the original coefficients and $0, \pm 1$. A solution obtained by nonzero minors bounds the coefficients of H_α and its representation simultaneously. The resulting H_α form a Gröbner basis, since their leading monomials generate the initial ideal. Thus no bound for coefficients of a previously chosen basis is presumed.

If the joint affine height of the input is H , a determinant of size s has height at most $sH + s \log s$. Normalizing a nonzero coordinate and concatenating polynomially many coefficient lists incurs only a polynomial factor. Consequently all these affine coefficient heights are at most $Q(D)(H + 1)$: the dependence on the logarithmic input height is linear. The same argument treats elimination and saturation; for example saturation by a is computed by eliminating z after adjoining $za - 1$.

For varieties, the degree and arithmetic Bézout inequalities give the following consequence: intersections, closures of coordinate projections, and choices of irreducible components have degrees polynomial in the input degrees and Chow heights at most a polynomial in the input degrees times $1 +$ the joint input height. Here and below the ambient dimensions are fixed. We use the arithmetic Bézout inequality [KPS01, Corollary 2.11], the projection bound [KPS01, Lemma 2.6], and additivity and nonnegativity of cycle heights [KPS01, Section 1.2.4]. The comparison of Chow coefficient height with this cycle height is [KPS01, Lemma 1.1, inequality (1.2), and Section 1.2.4]. One may work throughout with Chow forms, normalizing one nonzero coefficient to 1. The affine height of the normalized coefficient tuple then equals its projective height, by the product formula.

It is enough to recover equations *set-theoretically* from a Chow form. Indeed, if $Z \subset \mathbb{P}^N$ has dimension r and Chow form $\mathcal{C}_Z$, substitute the $r + 1$ hyperplanes

$$\sum_{j=1}^N T_{ij}(X_j - x_j X_0) = 0 \quad (0 \leq i \leq r)$$

in $\mathcal{C}_Z$ and take its coefficients as a polynomial in the T_{ij} . These coefficients vanish at x precisely when $[1 : x] \in Z$: outside Z , the hyperplanes through that point form a base-point-free system on Z , and $r + 1$ general members have empty intersection on Z . The

resulting equations have degree at most $(r + 1) \deg Z$ and controlled affine coefficient heights by expansion. Their number is polynomial in $\deg Z$.

Thus every variety used below admits an ideal $\mathfrak{b}$ of controlled generators with

$$V(\mathfrak{b}) = W, \quad \mathfrak{b} \subseteq I(W). \quad (3.4)$$

We need not compute generators of the radical $I(W)$. The distinction will be useful: ideal membership in $\mathfrak{b}$ suffices for evaluation on W , while nonvanishing on W will be imposed separately.

3.4 An ordinary-point zero estimate

The differential method compares successive derivatives in a finite-dimensional row space; see the classical multiplicity estimate of Bertrand and Beukers [BB85, Theorem 1]. We give the argument here because the algebraic variety on which the system is restricted varies in the descent, and the bound must be polynomial in its degree while allowing the second argument aR .

Lemma 3.2 (Uniform ordinary-point order bound). *For the fixed analytic data of Theorem 3.1 there is a polynomial $M(D)$ with the following property. Let W be an irreducible affine variety of degree at most D on which a is nonconstant. On a branch above the generic point of a component of $V_W(a)$, every allowed expression E which is not identically zero satisfies*

$$\text{ord}_{\mathfrak{p}} E < M(D) \text{ord}_{\mathfrak{p}} a. \quad (3.5)$$

The bound is independent of the heights of the coefficients of W and of the expression.

Proof. We first preserve the branch order on a curve of controlled degree, then use the ordinary differential equation to bound that order independently of coefficient heights.

Reduction to a curve. Write $k = \dim W$, and let $\tilde{D}$ be the normalization divisor defining the branch. At a general point p of $\tilde{D}$, both the normalization and $\tilde{D}$ are smooth, and the finite map from $\tilde{D}$ to its image D_0 is étale at p after restricting to dense open subsets, by separability in characteristic zero. Choose $k - 1$ general affine linear functions whose differentials restrict to a basis of the cotangent space of D_0 at the image of p . Their pullbacks restrict to a basis on $\tilde{D}$. Their common level set through p is consequently a smooth curve germ transverse to $\tilde{D}$ in the normalization. The downstairs linear section is proper and has total projective degree at most D . The transverse germ leaves the nonnormal locus, so its map to its image curve is generically the normalization isomorphism. It is thus the corresponding smooth germ on the normalization of an irreducible component of that linear section.

This slice preserves formal orders as well. If $\text{ord}_{\tilde{D}} a = e > 0$ and $\text{ord}_{\tilde{D}} E = N$, only the terms with $j \leq N/e$ contribute to the leading coefficient. After deleting a proper closed subset of $\tilde{D}$, write $a = t^e v$ and that finite truncation as $t^N u$, where t is a local equation of $\tilde{D}$ and u, v are units. On the transverse curve t is a uniformizer. Hence the orders are exactly e and N after restriction. All these conditions hold at a general slice point.

Let T_0 be the smooth projective normalization of this sliced curve. Its genus is polynomial in D , by a general birational plane projection and the arithmetic genus of a plane curve. Every ambient coordinate on T_0 is a quotient X_i/X_0 , so its polar divisor has degree at most D . In particular the maps defined by a and aR have degrees at most D and $2D$, respectively.

Adjoin the points on the fixed curves specified by the algebraic chart germs. Constant argument maps contribute constant chart factors and require no cover; in what follows index only the nonconstant argument maps. If the corresponding fixed maps to $\mathbb{P}^1$ have degrees d_i

and branch sets B_i , including infinity when it is a branch value, the chosen normalized fibre-product component $T \rightarrow T_0$ has degree at most $N_0 = \prod_i d_i$. It is étale outside $\bigcup_i \phi_i^{-1}(B_i)$, where ϕ_i is a or aR . This set has at most $\sum_i \#B_i \deg \phi_i = O(D)$ points, and the ramification contribution above each is at most N_0 . Thus Riemann–Hurwitz gives

$$2g(T) - 2 \leq N_0 \max\{0, 2g(T_0) - 2\} + O(D).$$

The degrees of the maps from T to the fixed curves are bounded by $N_0 \deg \phi_i$, so all pulled-back fixed divisors have polynomial degree. Poles of aR are included in this calculation through the fibre above infinity. All ramification indices are bounded as well; any further ramification multiplies the two orders in (3.5) by the same factor.

The ordinary differential system. Let $p \in T$ be the point of the specified branch. In direct sums and tensor products of the fixed pulled-back systems the expression takes the form

$$E = r_0 y,$$

where y is a horizontal column of a fixed rank N and r_0 is a rational row. Polynomial algebraic factors are absorbed in r_0 . A constant map to one of the fixed curves contributes a constant horizontal factor. The rational frames here are the pullbacks of the fixed frames; they are regular and invertible at p . In fact $a = 0$ and R is regular at the chosen affine divisor, so both arguments a and aR are at their ordinary centres. The connection and r_0 are therefore regular at p . Their global polar divisors have degrees polynomial in D , by the bounds on the degrees of the pullback maps and of the coordinate functions.

Choose a rational function z on T which is a uniformizer at p and whose polar divisor has degree at most $2g(T) + 1$. Such a function follows from Riemann–Roch by prescribing its first two jets at p and allowing a pole at another point. For $g(T) = 0$ a degree-one function suffices. Put $\delta = d/dz$, and write the horizontal equation as $\delta y = Ay$. The poles of the rational vector field δ are bounded in degree by the zero divisor of dz , hence by $2g(T) - 2 + 2 \deg z$. Differentiating a rational function adds at most its reduced polar divisor and the polar divisor of this vector field. It follows that A , r_0 , and any fixed number of their derivatives have polar degrees polynomial in D . They remain regular at p ; using a uniformizer rather than dividing by da accommodates arbitrary ramification of a .

Set

$$r_{i+1} = \delta r_i + r_i A.$$

Then $\delta^i E = r_i y$. The successive rows become dependent after at most N steps. If r_ρ is the first row dependent on its predecessors, their span $M = \langle r_0, \dots, r_{\rho-1} \rangle_{\mathbb{C}(T)}$ is stable under the dual connection: differentiating the dependence shows that every later row belongs to it. Here $\rho \leq N$.

Let $\mathcal{O} = \mathcal{O}_{T,p}$ and saturate the common kernel of these rows in $\mathcal{O}^N$. Saturation makes it a subbundle locally. It is stable under the ordinary connection, since its derivative is regular and belongs to the kernel generically. The quotient has an ordinary connection as well. Explicitly, if v is a local section annihilated by every row in M , stability of M under the dual connection gives $r(\nabla_\delta v) = 0$ for every $r \in M$. The ordinary connection makes $\nabla_\delta v$ regular, so it belongs to the saturated kernel. The kernel and its quotient are free over the local discrete valuation ring, and the induced quotient connection has no pole. The image $\bar{y}$ of y in this quotient is a nonzero horizontal section: a zero image would give $r_0 y = E = 0$. Conversely, if E were identically zero, all its derivatives $r_i y$ would be zero and y would lie in the kernel. The nonzero horizontal quotient section has nonzero reduction at p : the coefficient recurrence of an ordinary linear differential equation forces a horizontal section with zero initial value to be zero.

The integral dual of this free quotient is exactly the saturated row lattice $\mathcal{L} = M \cap \mathcal{O}^N$: its rows are the integral functionals annihilating the common kernel. Since $\bar{y}$ has nonzero reduction, one such functional $\lambda \in \mathcal{L}$ has λy a unit. Choose a nonzero $\rho \times \rho$ minor Δ of the matrix with rows $r_0, \dots, r_{\rho-1}$. All these rows are regular at p . Cramer's rule on the selected columns gives

$$\Delta \lambda = \sum_{i=0}^{\rho-1} c_i r_i, \quad c_i \in \mathcal{O}.$$

Hence at least one $i < \rho$ satisfies $\text{ord}_p(r_i y) \leq \text{ord}_p \Delta$. This is the elementary-divisor bound for the row lattice, expressed without choosing a new global frame. A nonzero rational function on T has equal total zero and pole degrees, so $\text{ord}_p \Delta$ is bounded by its polar degree, which is polynomial in D . Finally, since z is a uniformizer,

$$\text{ord}_p E \leq \text{ord}_p(\delta^i E) + i \leq \text{ord}_p \Delta + N - 1.$$

Increasing the resulting polynomial by 1 proves (3.5), since $\text{ord}_p a \geq 1$. This construction used only degrees, genera, ramification and fixed ranks; it used no arithmetic height. $\square$

3.5 The graph and its special fibre

Lemma 3.3 (Dimension of the constrained graph fibre). *Let W be an irreducible affine variety of dimension $k \geq 1$ with coordinate a nonconstant, let f be regular on W , and let $m \geq 1$ be an integer. Suppose*

$$\text{ord}_{\mathfrak{p}} f < m \text{ord}_{\mathfrak{p}} a$$

on every normalization branch over the generic components of $V_W(a)$. Let W' be the affine graph closure of $q = f/a^m$ over $W \setminus V(a)$. For every regular function g on W ,

$$\dim V(I(W'), a, q + g) \leq k - 2. \quad (3.6)$$

The set is empty when $k = 1$; the same convention applies if $V_W(a)$ is empty.

Proof. Write $A = \overline{\mathbb{Q}}[W]$, $u = f/a^m$, and let B be the finite normalization of A . The graph has coordinate ring $A[u]$, and $B[u]$ is finite over it. Suppose there were a point of the graph above a generic point $\mathfrak{p}$ of a component of $V_W(a)$. Lying over in $B[u]$ and then contraction to B give a prime $\mathfrak{r}$ above $\mathfrak{p}$. The local ring $B_{\mathfrak{r}}$ is a discrete valuation ring. The hypothesis says $u = v\pi^{-s}$ there, for a unit v , a uniformizer π , and $s > 0$. Its map into the local ring of the lifted graph point would satisfy $\pi^s u = v$ with π in the maximal ideal and v invertible, a contradiction.

The constructible image of $V_{W'}(a)$ therefore contains none of the generic points of $V_W(a)$. Each component of the latter has dimension $k - 1$, so the closure of this image has dimension at most $k - 2$. On imposing $q = -g$, projection from the graph to W is a closed immersion: in its coordinate ring the extra coordinate is already specified by g . Its image lies in the preceding constructible image, which proves (3.6). $\square$

3.6 Interpolation by successive powers of the parameter

The extra equation $q + g = 0$ in Lemma 3.3 is essential. For example, on $W = \mathbb{A}^2$ with coordinates (a, u) , take $f = u$ and $m = 1$. The graph $W' = \{aq = u\}$ still has a one-dimensional fiber over $a = 0$: its entire q -line maps to $(a, u) = (0, 0)$. Imposing $q + g_0(u) = 0$ leaves one point. Thus it is the projected boundary, and then the graph with this additional equation, that has dimension $k - 2$; the boundary of the graph alone need not have that dimension. Figure 1 displays this distinction.

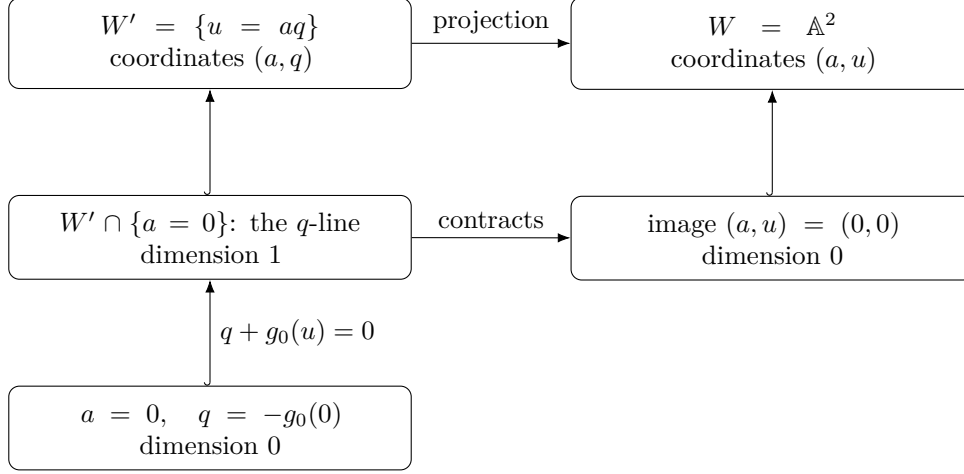


Figure 1: The graph $q = u/a$ over a two-dimensional affine base. Horizontal arrows are projections and vertical arrows are inclusions. The graph boundary is a line, but its image is a point. The additional equation $q + g_0(u) = 0$ selects one point on that line, giving the dimension $k - 2 = 0$ used in the layer count.

The next lemma gives a polynomial of low total degree with an arbitrarily large prescribed polynomial ratio of order to degree. Requiring nonvanishing on the variety is part of the construction.

Lemma 3.4 (Interpolation through successive layers). *Let $W' \subset \mathbb{A}^M$ be irreducible of dimension $k \geq 1$, with coordinates including a, U, q . Let $\mathbf{b} = (G_1, \dots, G_t)$ have controlled generators and $V(\mathbf{b}) = W'$. Suppose*

$$\dim V(\mathbf{b}, a, F_0) \leq k - 2, \quad F_0 = q + g_0(U),$$

and put $F_n = F_0 + \sum_{j=1}^{n-1} a^j g_j(U)$, where $m \geq 1$ is an integer and $\deg g_j \leq C(m + j + 1)$. Here M, C are fixed and all other complexities, including m , are polynomially bounded in d ; the coefficient heights through index n are bounded by $Q(n, d)\Psi(h)$.

For any prescribed positive integer r polynomially bounded in d , there are positive integers L and $n = rL$, both polynomially bounded in d , and a polynomial P such that

$$\deg P \leq L, \quad P \notin I(W'), \quad P = \sum_{\lambda=1}^t A_\lambda G_\lambda + AF_n + a^n B. \quad (3.7)$$

The degrees of all the coefficients in this representation are polynomial in d , and their joint affine height, including that of P , is at most $Q'(d)\Psi(h)$. The polynomial Q' may depend on the prescribed polynomial bound for r .

Proof. Fix a degree-compatible monomial order and a Gröbner basis of $\mathfrak{j} = (\mathbf{b}, a, F_0)$, together with representations of its elements in these displayed generators. Their degrees and heights have the bounds recorded above. For each reducible monomial, fix a basis element to divide by. Division in degree at most T can then be performed by a fixed decreasing sweep through all monomials of that degree range, allowing zero coefficients. Thus, for $\deg H \leq T$, it gives linearly in H

$$H = \text{rem}_{\mathfrak{j}} H + \sum_{\lambda} C_\lambda G_\lambda + aC + F_0 D. \quad (3.8)$$

There is a fixed bound Δ , polynomial in the complexities at this stage, such that every term on the right has degree at most $T + \Delta$. Indeed, degree-compatible division produces

quotient multipliers of degree at most $T - \deg H_\alpha$ for a basis element H_α . Substituting its fixed generator representation adds its excess degree, at most Δ ; it does not multiply T .

Work modulo $(\mathfrak{b}, F_n, a^n)$. In (3.8), the aC term advances one layer, while

$$F_0 D \equiv - \sum_{j=1}^{n-1} a^j g_j D$$

advances at least one layer. Starting with a polynomial of degree at most L , apply this division to each successive coefficient of $1, a, \dots, a^{n-1}$. The coefficient being divided at layer i has degree at most $L + B_0(i+1)$, for a polynomially bounded B_0 . To verify this induction, a jump by $j \geq 1$ adds at most $\Delta + C(m+j+1) \leq B_0 j$ to the degree. The sum of all jumps to layer i is i , and there is no substitution at the same layer. We retain every remainder and require it to be zero.

When $k = 1$, the dimension hypothesis makes $\mathfrak{j} = (1)$, so there are no remainder conditions. If $\mathfrak{j} \neq (1)$, then $k \geq 2$, and the number of standard monomials of degree at most T is at most

$$B_1(1+T)^{k-2},$$

with B_1 polynomially bounded. Here is an elementary uniform bound. If the monomial initial ideal has generators of degree at most D_1 , in a standard monomial at most $\dim V(\mathfrak{j})$ exponents can be at least D_1 . Otherwise its variables of large exponent would contain the support of a monomial generator, by the coordinate-prime description of the radical of a monomial ideal. Choose those variables and bound every other exponent by $D_1 - 1$. This gives the asserted bound with a constant depending polynomially on D_1 , since M is fixed. If $\mathfrak{j} = (1)$ there are no standard monomials and no conditions. Thus at most

$$B_1 n(1+L+B_0 n)^{k-2} \tag{3.9}$$

linear conditions suffice when the ideal is proper.

Choose k algebraically independent functions among the ambient coordinate functions on W' . Such a subset exists because these functions generate its function field. Let V_L be the span of their monomials of total degree at most L , regarded as actual polynomials in the ambient coordinates. Then

$$\dim V_L = \binom{L+k}{k}, \quad V_L \cap I(W') = 0.$$

This chooses independent polynomial representatives; no remainder map is presumed to descend canonically to classes modulo $I(W')$. One can equally extend V_L to a linear complement of $I(W') \cap \overline{\mathbb{Q}}[a, U, q]_{\leq L}$.

Set $n = rL$. For a proper $\mathfrak{j}$, the bound (3.9) is at most

$$B_2 r(1+B_0 r)^{k-2} L^{k-1}$$

for a polynomially bounded B_2 . A sufficiently large polynomial choice of L makes it smaller than $\binom{L+k}{k}$. In the unit-ideal case the conclusion holds already for any nonzero source vector. There is consequently a nonzero element $P \in V_L$ for which all remainders vanish. This proves both membership and $P \notin I(W')$.

For completeness, the construction also gives the asserted affine height bounds. Division in bounded degree is a linear operation on a coefficient space of polynomial dimension. During division the leading monomial strictly decreases among the monomials of degree at most T ; thus its coefficient transformation is a product of polynomially many matrices with

controlled affine heights. For composable matrices with inner dimension b , the coefficient-tuple convention gives

$$h(AB) \leq h(A) + h(B) + \log b.$$

Indeed, locally every entry is a sum of b products, and only infinite places contribute the factor b . The n layer operations therefore have polynomial affine heights. Their source basis consists of monomials, so has height zero. A kernel vector obtained by minors and normalized to have a nonzero coordinate equal to 1 has affine height $Q(d, n, L)\Psi(h)$.

The tracked divisions give a membership representation of bounded degree. Alternatively, apply the arbitrary-membership bound to $(\mathfrak{b}, F_n, a^n)$ and solve its coefficient equations with the chosen P : the degrees and number of unknowns are polynomial in d, n, L , and minors give the same affine height estimate. This bounds the scale of P, A_λ, A, B together. Since r, L, n are polynomial in d , (3.7) follows with the stated bounds. $\square$

3.7 Product formula and descent

Proof of Theorem 3.1. We descend through irreducible subvarieties $W \subseteq X$ containing z_* . At every step their degrees and degrees of chosen set-theoretic equations are polynomial in d , and the affine heights of those equations are at most $Q(d)\Psi(h)$. Initially take a component of X containing the point. The algebraic bounds above preserve these conditions under each operation below.

If a is constant on W , its constant value has height at most $Q(d)\Psi(h)$, by projection to the a -line and the component height bound. On the other hand the product formula and (3.2) give

$$h(a_*) = \sum_v w_v \log \max(1, |a_*|_v^{-1}) \geq S_{\text{small}} \geq h/Q_1(d). \quad (3.10)$$

Hence $h \leq Q(d)\Psi(h)$, which implies a polynomial bound in d : for fixed c , $1 + (\log h)^c \leq C_c \sqrt{h}$ for $h \geq 2$.

Suppose now that a is nonconstant and $k = \dim W \geq 1$. If a is nonconstant but $V_W(a)$ is empty, this case can be settled without an analytic equation. For the controlled ideal $\mathfrak{b}_W = (G_1, \dots, G_t)$, the ideal $(\mathfrak{b}_W, a)$ is the unit ideal. The membership and coefficient bounds provide

$$1 = \sum_i A_i G_i + aB$$

with polynomial degrees in d and joint affine height $Q(d)\Psi(h)$. At the target $1 = a_*B(z_*)$. Evaluating B on the selected embeddings costs only its coefficient height and the small local sizes of U_* , since $|a_*|_v < 1$. It follows that

$$S_{\text{small}} = \sum_{v \in \mathcal{V}} w_v \log |B(z_*)|_v \leq Q(d)\Psi(h).$$

Again $h \leq Q(d)\Psi(h)$ gives the required polynomial bound. We may therefore suppose that $V_W(a)$ is nonempty.

The total number of generic normalization branches above $V_W(a)$ is at most $\deg W$. Indeed take one general transverse linear slice meeting every component D_j of this fibre at general affine points. The curve components have total degree at most $\deg W$. Their normalizations contain the transverse germs above all these points. If a normalization divisor above D_j has residue degree f_i and $e_i = \text{ord}_i(a)$, it contributes $\deg(D_j)f_i e_i$ to the zero divisor of a on the normalized slice. That divisor has degree equal to the polar degree of a , at most $\deg W$. In particular the number of branches is bounded. This counts residue degrees and ramification multiplicities, even though the normalization itself is birational.

At each such branch choose the lowest nonzero order among the E_ν . To avoid cancellation of these leading terms in a combination, set

$$E = \sum_{\nu=1}^s t^{\nu-1} E_\nu$$

for a positive integer t . Each branch excludes at most $s - 1$ values of t , since its leading-coefficient polynomial is nonzero. There is therefore a permissible t of polynomial size in d . This combination preserves the coefficient and tail bounds: at finite places its coefficients are integral, and the extra triangle-inequality factor at infinite places has total logarithm polynomial in d . Absorb it in the b_v . Write $E = \sum a^j e_j(U)$.

By [Lemma 3.2](#), an integer m polynomially bounded in d can be chosen so that, for

$$f = \sum_{j=0}^{m-1} a^j e_j(U),$$

one has $\text{ord}_{\mathfrak{p}} f < m \text{ord}_{\mathfrak{p}} a$ on every relevant branch. Indeed the omitted terms have order at least $m \text{ord}_{\mathfrak{p}} a$, since all their coefficients are regular on the affine variety. Form the graph W' of $q = f/a^m$. Its equations with controlled support are obtained by saturating $(\mathfrak{b}_W, a^m q - f)$ by a ; the closure of its support is exactly the graph. This takes place in one more variable and has polynomial degree and affine height bounds.

The target lifts to $z'_* = (a_*, U_*, q_*)$. Its global height is bounded using the rational expression for q_* :

$$h(z'_*) \leq Q_5(d)h + Q_6(d)\Psi(h). \quad (3.11)$$

Here Q_5 is fixed at this stage, before the interpolation degree and order are chosen. Its selected local size must instead be estimated using the exact equation $E(z_*) = 0$. The tail bound gives

$$|q_*|_v = \left| a_*^{-m} \sum_{j \geq m} a_*^j e_j(U_*) \right|_v \leq \exp((m+1)b_v). \quad (3.12)$$

In particular the sum of the positive local logarithmic sizes of (U_*, q_*) is at most $Q_7(d)\Psi(h)$. The new equation and its finite truncations are

$$F = q + \sum_{j \geq 0} a^j e_{m+j}(U), \quad F_n = q + \sum_{j=0}^{n-1} a^j e_{m+j}(U).$$

At the target,

$$|F_n(z'_*)|_v \leq |a_*|_v^n \exp((m+n+1)b_v). \quad (3.13)$$

By [Lemma 3.3](#), $V(I(W'), a, q + e_m)$ has dimension at most $k - 2$. For a controlled ideal $\mathfrak{b}$ defining W' set-theoretically, $V(\mathfrak{b}, a, q + e_m)$ is the same set and has the same dimension. Apply [Lemma 3.4](#), with the ratio $r = n/L$ to be chosen shortly. At the lifted target its identity becomes

$$P(z'_*) = A(z'_*)F_n(z'_*) + a_*^n B(z'_*).$$

Equations (3.1), (3.12), and (3.13), together with the coefficient and membership degree bounds, imply

$$\sum_{v \in \mathcal{V}} w_v \log |P(z'_*)|_v \leq -nS_{\text{small}} + Q_8(d, n, L)\Psi(h). \quad (3.14)$$

All positive contributions here are sums of affine coefficient heights, positive local coordinate sizes, the b_v , and infinite-place triangle-inequality terms. There is no error proportional to the

number of finite places. At the remaining places evaluate P itself, rather than its membership representation. Since $\deg P \leq L$, (3.11) gives

$$\sum_{v \notin \mathcal{V}} w_v \log |P(z'_*)|_v \leq LQ_5(d)h + Q_9(d, n, L)\Psi(h). \quad (3.15)$$

The choices can now be made in the required order. The graph and m have already fixed Q_5 . Choose an integer $r > 2Q_1(d)(Q_5(d) + 1) + 1$, polynomially bounded in d . Next choose the polynomially large L supplied by Lemma 3.4, and set $n = rL$. If $P(z'_*) \neq 0$, its product formula and (3.14)–(3.15) give

$$0 \leq \left(-\frac{n}{Q_1(d)} + LQ_5(d) \right) h + Q_{10}(d, n, L)\Psi(h).$$

The coefficient of h is at most $-L$. As n, L are now fixed polynomials in d , this inequality again implies $h \leq Q(d)\Psi(h)$ and hence a polynomial height bound. Thus, beyond a polynomial threshold chosen *after* m, r, L, n , the value $P(z'_*)$ must be zero.

Since $P \notin I(W')$, the intersection $W' \cap V(P)$ has dimension at most $k - 1$. Project it to the original (a, U) -space, take its closure, and choose an irreducible component containing z_* . Such a component exists because $a_* \neq 0$ and the graph projection is an isomorphism over that open set. Projection cannot increase dimension. The degree and arithmetic bounds above preserve the induction hypotheses. Discard the graph coordinate q ; at the next step construct a new graph from the original common equations on this smaller variety. Consequently the ambient number of variables stays fixed.

The dimension drops at every continuing step, so there are at most $\dim X$ such steps. A zero-dimensional final variety has constant a and is covered by (3.10). Taking the maximum of the finitely many polynomial thresholds proves the theorem. All component and coefficient field extensions made during the proof extend the selected local data by all embeddings, as specified above; they do not alter any smallness sum or introduce a field-degree factor. $\square$

4 Arithmetic coordinates and common local relations

We construct the algebraic coordinates and common local equations needed for Theorem 3.1. The coordinates will have controlled global height and much smaller norms on the center disks. A weighted proximity estimate then ensures that one common system of equations retains enough smallness of the local parameter. Throughout this section the curve, its finite cover, its algebraic frames, and the center are fixed. In particular the conclusions apply after any one fixed auxiliary level cover. No condition on the reduction type of the center is imposed.

Write $\pi : \mathcal{A} \rightarrow S$ for the abelian scheme and put $\mathcal{H} = R^1\pi_*\Omega_{\mathcal{A}/S}^\bullet$. We use the operator realization of $\mathcal{V} \subset \text{End}(\mathcal{H})$ from Lemma 2.2. On actual endomorphisms, Tr denotes the trace on first cohomology, so that $\text{Tr}(1) = 4$. As in (2.2), $\langle u, v \rangle = \frac{1}{4} \text{Tr}(uv)$. For an endomorphism u , set

$$q_\lambda(u) = \text{Tr}(uu^\dagger),$$

where $\dagger$ is the principal Rosati involution. This is a positive definite integral quadratic form on $\text{End}(A)$.

For a number field F containing coordinates under consideration, our weights are $w_v = [F_v : \mathbb{Q}_v]/[F : \mathbb{Q}]$, with the usual unsquared absolute values. Thus $h(z) = \sum_v w_v \log \max(1, \|z\|_v)$ for an affine tuple z , where the norm is the coordinate maximum. The same sums can be written over embeddings into $\mathbb{C}$ or $\mathbb{Q}_p$, giving each embedding weight $1/[F : \mathbb{Q}]$. If a selected set of embeddings is extended to a larger field, we always take *all* extensions. Their weights

sum to the old weight. This convention also permits grouping embeddings in a normal closure without charging its degree to any estimate.

After omitting finitely many points, fix a rational frame of $\mathcal{V}$ regular at t_0 , affine coordinates z on a neighborhood of t_0 , and $x \in K(S)$ with a simple zero at t_0 . The inverse branch through t_0 is denoted $z = z(X)$, with $z(0) = z(t_0)$. Shrink the neighborhood so that these data and the connection are regular there. Choose two independent integral trace-free Rosati-symmetric endomorphisms of A_{t_0} as in [Lemma 2.2](#), enlarge K once to define them, and denote their de Rham coordinate pair by $B = (B_1, B_2)$. For ordered vector pairs $\mathbf{v} = (v_1, v_2)$ and $\mathbf{w} = (w_1, w_2)$, write $\langle \mathbf{v}, \mathbf{w} \rangle = (\langle v_i, w_j \rangle)_{i,j=1}^2$. Put $a = x(t) \neq 0$, $h = \max\{2, h_H(t)\}$ for a fixed nonnegative ample logarithmic height on $\overline{S}$, and $d = \max\{2, [K(t) : K]\}$. All constants below depend only on the fixed data.

4.1 Integral endomorphisms and de Rham coordinates

Lemma 4.1 (Potential good reduction). *An abelian surface whose geometric endomorphism algebra is a quaternion division algebra over $\mathbb{Q}$ has potentially good reduction at every finite place.*

Proof. Fix a finite place. The semistable reduction theorem [[BLR90](#), Section 7.4, Theorem 1], followed by a further finite extension, gives semistable reduction and defines all geometric endomorphisms. They extend to the Néron model and preserve the torus in its connected special fiber. If that torus has positive rank r , its rational character space is a nonzero unital module, on one side or the other, for the division algebra $D = \text{End}^0(A_{\overline{K}})$. Every finite dimensional module over a division algebra is free over it, so its rational dimension is a positive multiple of $\dim_{\mathbb{Q}} D = 4$. This contradicts $r \leq 2$. The toric rank is therefore zero, and the semistable model is an abelian scheme. The extension used here is local; no uniform bound on its degree is needed for an absolute height computation. $\square$

Proposition 4.2 (Small labeled integral vectors). *There are constants $c, C > 0$ with the following property. For every target t , an extension $L/K(t)$ of degree at most C defines all geometric endomorphisms of A_t . There are two independent endomorphisms $u_1, u_2 \in \text{End}_L(A_t)$ which are symmetric and trace free, span E_t , and satisfy*

$$q_{\lambda_t}(u_j) \leq C(dh)^c \quad (j = 1, 2). \quad (4.1)$$

The corresponding labeled vectors are denoted $P = (P_1, P_2)$ in de Rham coordinates. Conjugating a labeled target always means conjugating these endomorphisms as well. This pair need not generate the saturated integral cohomology lattice of E_t .

Proof. The Galois action on the geometric endomorphism lattice has finite image in $\text{GL}_r(\mathbb{Z})$, with $r \leq 16$. Indeed a finite set of lattice generators is defined over a finite extension. A finite subgroup of $\text{GL}_r(\mathbb{Z})$ injects into $\text{GL}_r(\mathbf{F}_3)$, because the kernel of reduction modulo 3 is torsion free. Its order is consequently bounded in terms of r alone. The fixed field of the kernel gives the asserted L .

For a principally polarized abelian variety of fixed dimension, the endomorphism estimate of Masser–Wüstholz gives a rational spanning set of integral endomorphisms whose Rosati lengths are bounded by a fixed power of

$$\max\{2, [L : \mathbb{Q}], h_F(A_t)\}.$$

Here polynomial dependence on the field degree is essential. It is explicit in the Proposition of [[MW94](#), Section 4], with its parameter m equal to the dimension of the abelian variety; the alternative involving a smaller image is then zero. Its lattice Lemma 2.1 also gives a

basis with the same kind of bound. See also [DO22, Theorem 6.6 and Section 6.7] for the polynomial dependence in this application. The theta-height comparison [Paz12, Theorem 1.1 and Corollary 1.3], on one fixed theta level, and functoriality of heights on a fixed curve give

$$h_F(A_t) \leq Ch.$$

The degree of passage to that fixed level is bounded.

For clarity, the projection to the required plane can be made without target-dependent denominators. The map

$$u \longmapsto 4(u + u^\dagger) - \text{Tr}(u + u^\dagger)1$$

preserves integrality, has symmetric trace-free image, and increases the Rosati norm by at most a fixed factor. By Lemma 2.2 its rational image has dimension two. Apply this map to the small spanning set and select two independent images. This proves (4.1). The norm and trace assertions are unchanged under field conjugacy: they are traces of actual endomorphisms and their Rosati adjoints. $\square$

The next distinction is useful. A rational frame can have large denominators at a moving point, even when the endomorphism is integral. Their global cost is linear in h . On fixed sufficiently small neighborhoods of the center the frames are uniformly controlled, and the restricted cost is much smaller.

Proposition 4.3 (Coordinate heights and local norms). *Choose the vectors of Proposition 4.2. Their affine coordinates in the fixed frame satisfy*

$$h(z(t), a, P) \leq C(h + \log d). \quad (4.2)$$

There are center neighborhoods at every place, with a single integral description outside a fixed finite set of rational primes, such that for any selected weighted set $\mathcal{V}$ on the actual center branches,

$$\sum_{v \in \mathcal{V}} w_v \log \max(1, \|z(t), P\|_v) \leq C(1 + \log h + \log d). \quad (4.3)$$

These statements hold for the finitely many conjugate fixed charts. They also hold after a bounded denominator and a polarized isogeny of any one fixed multiplier are applied to the vectors.

Proof. Global finite-place bounds. We first bound the denominators of integral endomorphisms in the rational frame. Potential good reduction supplies an integral de Rham lattice at each target; a fixed ideal measures the denominators of the frame and its dual relative to that lattice. Pass, for this argument only, to a fixed common cover carrying two coprime fine levels $N_1, N_2 \geq 3$. The integral fine moduli scheme $\mathcal{A}_2(N_i)$ is quasi-projective over $\mathcal{O}_K[1/N_i]$ [Cha85, Chapter I, Theorem 1.4 and Section 1.7]. Embed it in projective space over this ring, take its scheme-theoretic closure, and then take the closure of the graph of the curve map in the product with the fixed projective curve model. Denote this proper graph model by X_i . Take its projective closure over $\mathcal{O}_K$ when using global model heights. The inverse image U_i of the abelian interior carries the integral vector bundle $\mathcal{H}_i$ of first de Rham cohomology and its dual. At a prime $p \nmid N_i$, a target acquires good reduction by Lemma 4.1; its prime-to- p level extends after a finite local extension. Properness extends the point to X_i . The good-reduction abelian scheme with prime-to- p level also supplies a valuation-ring point of the abelian interior. Its map to the projective closure agrees with the graph map on the generic fiber, hence everywhere by separatedness. The moduli image of the extended graph point therefore lies in the abelian interior. Consequently the extended point belongs to U_i .

This last assertion concerns the moduli image. The point is allowed to reduce above a boundary point of the original curve. Thus we use U_i , rather than a preassigned integral open of the curve, to compare lattices. The two choices of i cover every residue characteristic.

Choose a rational frame $e_1, \dots, e_4$ of $\mathcal{H}$ such that it and its dual $e^1, \dots, e^4$ are regular at t_0 . Form on U_i the coherent denominator ideals

$$\mathcal{I}_+ = \{f : fe_j \in \mathcal{H}_i \text{ for all } j\}, \quad \mathcal{I}_- = \{f : fe^j \in \mathcal{H}_i^\vee \text{ for all } j\}.$$

These are coherent colon ideals, and extend to coherent ideals on the noetherian projective model X_i [The26, Lemma 28.23.1(2), Tag 01PE]. Their product, further multiplied by a denominator ideal for the fixed rational conversion from operator entries to our frame of $\mathcal{V}$, gives a fixed ideal $\mathcal{J}_i$. Multiplication by $\mathcal{J}_i$ makes all the specified sections integral on U_i . Remove its finitely many generic zeros from the curve. No duality assertion about arbitrary coherent extensions across the boundary is required. For a lifted target over a valuation ring R , write $\mathcal{J}_i(t)R = (\eta)$ and $\Lambda = H_{\text{dR}}^1(\mathcal{A}_t/R)$. The frame vectors belong to $\eta^{-1}\Lambda$ and their duals belong to $\eta^{-1}\Lambda^\vee$. An actual integral endomorphism preserves Λ : it extends to the abelian scheme [BLR90, Section 1.2, Proposition 8], and de Rham cohomology is functorial. Each of its matrix entries is therefore in $\eta^{-2}R$. The extra fixed conversion of coordinates gives, for some fixed integer b ,

$$\log \max(1, \|P\|_v) \leq b\lambda_{\mathcal{J}_i, v}(t) + c_v \quad (v \nmid N_i), \quad (4.4)$$

where $c_v = 0$ outside finitely many fixed primes.

Here is explicitly why the right side has the claimed global cost. For a sufficiently ample fixed line bundle $\mathcal{L}$ on X_i , $\mathcal{J}_i\mathcal{L}^n$ is generated by finitely many fixed sections $s_0, \dots, s_r$ [The26, Proposition 28.27.13(7), Tag 01Q3]. Choose s_k with $s_k(t) \neq 0$. With the model norm at a finite place, $\lambda_{\mathcal{J}_i, v}(t) \leq -\log \|s_k(t)\|_v$. The product formula for this nonzero element of the metrized line $\mathcal{L}_t^n$ gives

$$\sum_{v \nmid \infty} w_v \lambda_{\mathcal{J}_i, v}(t) \leq nh_{\mathcal{L}}(t) + \sum_{v \mid \infty} w_v \log \|s_k(t)\|_v \leq nh_{\mathcal{L}}(t) + C.$$

The last bound uses the bounded sup norms of the finitely many fixed sections on the fixed projective complex model. The finite ideal heights are nonnegative, so this also bounds any restricted finite sum, such as the sum over $p \nmid N_i$. Every generic projective curve and map used here is fixed, so $h_{\mathcal{L}}(t) = O(h)$. Equation (4.4), summed using the two models, bounds all finite denominator contributions by $O(h)$, independently of the local extension used for good reduction.

Global archimedean bounds. At an archimedean place the Hodge norm of an actual endomorphism is $q_{\lambda_t}^{1/2}$, up to a fixed convention. In a rational de Rham frame, the norms of the frame and dual have at most a fixed power of the inverse distance to the finitely many omitted points, times powers of its logarithm. The algebraic Gauss–Manin connection is regular singular by [Kat70, Theorem 14.1]; this property passes to its duals, tensor products and subquotients [Kat70, Proposition 11.3 and Remark 11.3.4]. After a fixed ramified cover at each puncture, compare its algebraic logarithmic lattice with the canonical extension of the variation. Both frames and their inverses differ by meromorphic matrices with finite pole orders. Schmid’s norm estimates for the variation and its dual [Sch73, Theorems 4.9 and 6.6’] then give the asserted power and logarithmic bounds for the rational frame and its dual. The sum of the positive logarithms of these factors is $O(h)$ by the local height inequalities for the fixed omitted divisor. The contribution of (4.1) is $O(\log h + \log d)$. Affine base coordinates and a are fixed rational functions and have height $O(h)$. This proves (4.2).

Bounds on the center disks. For (4.3), outside finitely many primes the family, parameter chart, frame, and dual are integral near the center, with the frame invertible. All integral

endomorphism coordinates then have norm at most one on that residue neighborhood. At a fixed exceptional prime p , extend the local coefficient field once so that the fixed center has good reduction and a chosen full M -level structure, where $M \geq 3$ and $p \nmid M$. Adding this level to the original characteristic-zero curve gives a finite étale cover; at the chosen lift of the center, the p -adic inverse function theorem gives an analytic section over a sufficiently small center disk. Every sufficiently close target on the original branch therefore has a lift on this section after coefficient extension, including when the original level is divisible by p . Forgetting the original level maps the lifted disk to the prime-to- p fine moduli scheme. On a smaller closed disk its coordinates lie in an integral open about the fixed center and reduce to the same special point. An integral de Rham frame on this moduli open pulls back to a frame whose change-of-basis matrix with the original frame, and its inverse, are analytic and bounded on the disk. Thus both the finite-place comparison cost and the base-coordinate norms are bounded by fixed local constants.

At infinity choose compact disks within the regular charts; the frame and dual Hodge norms are bounded there. Only $O(\log h + \log d)$ remains from the Rosati norm. The sum of weights above a fixed rational prime is one, as is the sum of archimedean weights, which proves the restricted sum.

Finally a polarized isogeny of fixed multiplier carries Rosati norms isometrically under conjugation on operators. Its inverse introduces only a bounded denominator into an actual endomorphism. This changes the finite estimates at finitely many fixed primes by fixed constants, and proves the last assertion. $\square$

4.2 Horizontal comparison on good-reduction disks

We next express pairings between centered and target endomorphisms in one cohomology space. At finite places the common special fiber supplies that space, and functorial comparison must apply to each individual endomorphism even when it does not extend over the base curve. Let $Y(X)$ be the rank-five horizontal transport *back* to the center, normalized by $Y(0) = I$. It is obtained from the corresponding transport on $\mathcal{H}$ by the operator construction. Its entries are ordinary-point Taylor series over K .

Proposition 4.4 (Functorial good-reduction comparison). *At a finite place, suppose a point t is on a sufficiently small actual center branch. After coefficient extension the two polarized abelian schemes have a common special fiber $\bar{A}$. More precisely, over a finite local extension F with residue field k , put $F_0 = W(k)[1/p]$ and $D = H_{\text{cris}}^1(\bar{A}/W(k))[1/p]$. There are rational comparison isomorphisms*

$$c_u : H_{\text{dR}}^1(A_u/F) \longrightarrow D \otimes_{F_0} F, \quad u = t_0, t,$$

whose transition $c_{t_0}^{-1} c_t$ is horizontal transport. They are compatible with morphisms of either lifted fiber and their specializations, including morphisms which do not extend over the base curve. Ramification of the coefficient field is allowed.

In particular, if β_i and α_j are the specializations of the centered and target endomorphisms, then

$$\langle B_i, Y(a)P_j \rangle = \frac{1}{4} \text{Tr}(\beta_i \alpha_j) \in \mathbb{Q} \quad (1 \leq i, j \leq 2). \quad (4.5)$$

The cross pairings belong to $\frac{1}{4}\mathbb{Z}$, and their ordinary absolute values are at most $C(dh)^c$. At archimedean places the same assertions hold with integral Betti transport in place of specialization.

Proof. Work on a smooth integral prime-to- p fine moduli chart $\mathfrak{M}$ over a Witt base, and let z be the common special point. The relative first crystalline cohomology of the universal abelian

scheme defines a convergent isocrystal $\mathcal{E}$ on the special fiber of this chart. Its value at z is the crystalline cohomology of the actual abelian variety $\overline{A}$ over z . The canonical realization of $\mathcal{E}$ on any smooth proper lifting is its relative de Rham cohomology with Gauss–Manin connection; see [MP12, Theorems 4.16 and 4.26], the latter using [Ogu84, Theorem 3.8.2].

Here is the common evaluation explicitly. Over the local field F in the statement, put $\mathfrak{T} = \mathrm{Spf} \mathcal{O}_F\langle Z \rangle$ and substitute $X = aZ$ into the chosen inverse branch. Shrinking the fixed center disks ensures that the resulting chart coordinates belong to $\mathcal{O}_F\langle Z \rangle$ and differ from their centered values by elements of $\mathfrak{m}_F \mathcal{O}_F\langle Z \rangle$, where $\mathfrak{m}_F$ is the maximal ideal of $\mathcal{O}_F$. Thus the map to $\mathfrak{M}$ reduces constantly to z on $(\mathfrak{T} \bmod p)_{\mathrm{red}} = \mathrm{Spec} k[Z]$. This remains true when F is ramified and $a \notin p\mathcal{O}_F$: the image of $\mathfrak{m}_F$ in $\mathcal{O}_F/p$ is nilpotent. Consequently $\mathfrak{T}$ is an enlargement of z , and the crystal transition for its structural map to the point enlargement gives a canonical horizontal identification

$$\mathcal{E}_{\mathfrak{T}} \simeq D \otimes_{F_0} \mathcal{O}_{\mathfrak{T}}[1/p].$$

This is the constant-reduction construction of [MP12, Definitions 4.10–4.13 and Section 4.7].

At $Z = 0$ and $Z = 1$, the relative comparison gives precisely the two maps c_{t_0} and c_t of the statement. Their compatibility with restriction to a fiber identifies them with the functorial absolute crystalline–de Rham comparisons; an explicit absolute statement with arbitrary ramification is [KP22, Appendix B, Proposition B.4]. Proper formal GAGA identifies completed and ordinary de Rham cohomology. An endomorphism of either individual fiber extends over its good-reduction abelian scheme and is carried to its actual specialization on $\overline{A}$. This argument needs only that individual endomorphism, without an extension to neighboring fibers. The common fiber is $\overline{A}$ itself, with the same specialization maps and prime-to- p markings as in smooth proper base change. The construction does not introduce an auxiliary isogeny between special fibers.

The crystal identification is horizontal and is the identity when the two evaluations coincide. Therefore $c_{t_0}^{-1}c_t$ is the identity-normalized horizontal transport, and uniqueness for the ordinary differential equation identifies its operator realization with $Y(a)$. The resulting relative matrix identity is also discussed in [Pap25a, Theorem 3.4 and Remark 3.6].

Outside finitely many fixed primes our curve chart and connection are integral; the Taylor comparison converges, including over ramified extensions, whenever

$$-\log |a|_v > \frac{2 \log p}{p-1}. \quad (4.6)$$

Indeed the divided-power terms have denominators dividing the corresponding factorial. At the finitely many other primes, Lemma 4.1 for the fixed center and a prime-to-characteristic moduli chart give the same assertion on a fixed smaller disk.

Functoriality identifies both operator realizations with the operators β_i, α_j on the same crystalline vector space, proving (4.5). The characteristic polynomial of an endomorphism of an abelian variety has integral coefficients and is independent of the Weil cohomology: it is characterized by the degree polynomial $\deg(n-u)$ [Mil86, Section 12]. For the crystalline realization used here, the cohomological assertion can be seen directly. Rational crystalline cohomology is a Weil cohomology [MP12, Remark 4.15]. Cup product identifies the cohomology of the abelian special fiber with the exterior algebra on H^1 : comparison with its good lift reduces this to the corresponding statement for a complex abelian variety. Thus $[n]-u$ acts on top cohomology by $\det(n-u \mid H^1)$; the degree formula identifies that action with $\deg([n]-u)$. Equality for all sufficiently large integers n identifies the characteristic polynomials, and hence their traces. This applies also to $\beta_i \alpha_j$, so its trace is integral and the pairing in (4.5) belongs to $\frac{1}{4}\mathbb{Z}$. Specialization preserves the polarization, Rosati involution, and these trace

polynomials. Positivity of the Rosati form in characteristic p [Mil86, Theorem 17.3] gives

$$|\mathrm{Tr}(\beta_i \alpha_j)|^2 \leq \mathrm{Tr}(\beta_i^2) \mathrm{Tr}(\alpha_j^2) \leq C(dh)^c.$$

At infinity parallel transport identifies integral first cohomology lattices, so the cross pairing again belongs to $\frac{1}{4}\mathbb{Z}$. On the compact center disk the Hodge norms at the two points are uniformly comparable under this transport. Cauchy–Schwarz for the resulting positive norm and (4.1) give the same size bound. Notice that the two transported operators need not both be Hodge endomorphisms at the center in this last argument. $\square$

4.3 Smallness on the actual center branches

Lemma 4.5 (Proximity and the factorial cutoff). *Use the fixed center disks of Propositions 4.3 and 4.4, and at every remaining prime impose (4.6). Let $\mathcal{V}_0(t)$ be all weighted embeddings at which t belongs to these actual branches. Then*

$$\sum_{v \in \mathcal{V}_0(t)} w_v \log(1/|a|_v) \geq ch - C(1 + \log^2(dh)). \quad (4.7)$$

In particular the sum is at least $ch/2$ if $h \geq Q_0(d)$ for a fixed polynomial Q_0 .

Proof. Use the effective divisor $D_0 = [t_0]$ on the smooth completion $\bar{S}$. Since it has positive degree, some fixed multiple of D_0 minus the ample divisor defining h is ample. The height of an ample divisor is bounded below, and hence

$$h_{D_0}(t) \geq ch - C.$$

Take local Weil functions for this divisor. On the actual branch through t_0 they equal $-\log|x(t)|_v$ up to fixed bounded errors. Outside the chosen branch neighborhoods their positive contributions are bounded; the error is zero at all but finitely many primes. At an unexceptional prime the integral parameter chart is étale at the center, so the branch is precisely the Henselian inverse branch with that reduction. Other zeros of the rational function x are not counted. Before the factorial cutoff, the weighted proximity sum is therefore at least $ch - C$.

Here is a uniform bound for the loss caused by that cutoff. Put $D = [K(a) : \mathbb{Q}] \leq Cd$ and $H = h(a) \leq Ch$. If a rational prime p has any place where $|a|_v < 1$, it contributes at least $(\log p)/D$ to $h(a^{-1}) = H$: writing a local valuation in integral units cancels its ramification index against the absolute weight. For the set $\mathcal{P}(a)$ of these primes,

$$\sum_{p \in \mathcal{P}(a)} \log p \leq DH, \quad \#\mathcal{P}(a) \leq DH/\log 2. \quad (4.8)$$

Write $\alpha_p = \log p/(p-1)$. At places over p discarded by (4.6), the total weighted loss is at most $2\alpha_p$, since their weights sum to at most one. For any set of N primes,

$$\sum \alpha_p \leq C \sum_{k=1}^N \frac{\log(k+1)}{k} \leq C(1 + \log^2(N+1)).$$

Here $\log x/(x-1)$ decreases for $x > 1$, and the k th smallest prime is at least $k+1$. Apply this with (4.8). Restricting the finitely many exceptional and archimedean disks costs only a fixed further constant. This proves (4.7); its last assertion follows by elementary absorption of the logarithmic term for $h \geq Q_0(d)$. $\square$

4.4 Grouping the equations

Include in the fixed exceptional set all denominator primes of the centered vectors and the trace pairing, in all fixed conjugate charts. At infinity and at the finitely many exceptional primes, only $O(d)$ embeddings of the labeled field occur, so we can keep the rational cross-trace matrix as a coefficient of each group's equations. The number of remaining primes can grow with the target. There we compare a trace equation with its image under a lift of residue Frobenius. The rational trace is fixed, so subtraction eliminates it; only $O(d)$ conjugates of the complete labeled tuple can occur. Retaining the labels also records Frobenius's action on the actual endomorphisms, as needed in [Section 5](#).

Proposition 4.6 (Common relations with controlled coefficients). *Suppose $h \geq Q_0(d)$, increasing Q_0 if necessary. The proximity embeddings can be partitioned into at most $Q_1(d)$ groups, and one group $\mathcal{V}$ has*

$$S_{\text{small}} = \sum_{v \in \mathcal{V}} w_v \log(1/|a|_v) \geq h/Q_2(d). \quad (4.9)$$

On each group there is one fixed list of analytic equations, evaluated at one algebraic tuple (a, U) , of one of the following forms:

$$\langle B, Y(a)P \rangle = T, \quad \text{with } T \in M_2(\mathbb{Q}), \quad (4.10)$$

$$\langle B, Y(a)P \rangle = \langle B', Y'(aR)P' \rangle, \quad R = b/a, \quad b = x'(t'), \quad (4.11)$$

together with the actual branch equations

$$z - z(a) = 0, \quad z' - z'(aR) = 0 \quad \text{when a second chart is present.} \quad (4.12)$$

The primed data are one simultaneous field conjugate of the complete labeled unprimed data. In (4.11) all places are finite and $|R|_v = 1$ on the group. The Frobenius of the common reduction transports each of the two labeled pairs to its primed pair.

The tuple has degree at most $Q_3(d)$ over K , and

$$h(a, U) \leq Q_3(d)h, \quad \sum_{v \in \mathcal{V}} w_v \log \max(1, \|U\|_v) \leq Q_3(d)(1 + \log h). \quad (4.13)$$

Writing the common list as $E(a, U) = \sum_{j \geq 0} a^j e_j(U)$, it has fixed length and

$$\begin{aligned} \deg e_j &\leq C(1 + j) & (j \geq 0), \\ h(\text{affine coefficient tuple of } (e_0, \dots, e_l)) &\leq Q_4(l, d)(1 + \log h) & (l \geq 0). \end{aligned} \quad (4.14)$$

There are $b_v \geq 0$ with

$$\left\| \sum_{j \geq l} a^j e_j(U) \right\|_v \leq |a|_v^l \exp((1 + l)b_v) \quad (l \geq 0), \quad (4.15)$$

$$\sum_{v \in \mathcal{V}} w_v b_v \leq Q_5(d)(1 + \log^2 h). \quad (4.16)$$

The series are formed from fixed ordinary-point algebraic connection systems in a and aR , and fixed algebraic chart series, by polynomial operations of fixed degree.

Proof. First choose a field L defining the complete labeled tuple: the point, its two endomorphisms, and the fixed curve, center, and frames. By [Proposition 4.2](#), $[L : \mathbb{Q}] \leq Cd$. Let M be

a normal closure. We do not bound $[M : \mathbb{Q}]$. Compute weights by extending every selected embedding to *all* embeddings of M ; every old weighted sum is preserved.

At infinity or a fixed exceptional prime, the four entries of the cross trace are the rational constants of [Proposition 4.4](#). At any one such place type, an embedding of the labeled field L into its local algebraic closure determines the Taylor evaluation and the constant. There are at most $[L : \mathbb{Q}]$ possibilities. Group by these embeddings, by the finitely many fixed charts, and then by the constant matrix. This gives $O(d)$ groups of the form (4.10), with $h(T) \leq C(1 + \log d + \log h)$. We count the constants actually obtained from these embeddings; we do not count all rational matrices of that height.

At an unexceptional prime p , for each embedding $\rho : M \hookrightarrow \overline{\mathbb{Q}}_p$, choose $\sigma \in \text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ inducing the p th-power automorphism of $\overline{\mathbb{F}}_p$. Apply σ to (4.5). The trace on the right is rational, and is fixed by σ . Thus

$$\langle \rho B, Y^\rho(\rho a)\rho P \rangle = \langle \sigma \rho B, Y^{\sigma \rho}(\sigma \rho a)\sigma \rho P \rangle.$$

On L , the embedding $\sigma \rho$ equals $\rho \tau$ for one of the at most $[L : \mathbb{Q}]$ embeddings $\tau : L \rightarrow M$. Partition by τ and the fixed conjugate charts. Define $(B', P', t', b) = \tau(B, P, t, a)$ in this group. This gives the same analytic equation (4.11) at every embedding of the group. The compositum $L\tau(L)$ has degree at most $[L : \mathbb{Q}]^2$, so the field defining the coordinates of this equation has polynomial degree even though M need not.

A local automorphism preserves the absolute value, whence $|b|_v = |a|_v$ and $|R|_v = 1$. Globally $h(R) \leq h(a) + h(b) \leq Ch$. On the common special fiber, relative Frobenius $F : \overline{A} \rightarrow \overline{A}^{(p)}$ satisfies $F\alpha = \alpha^{(p)}F$ for every endomorphism α , and similarly for the centered endomorphisms. Thus conjugation by F , which is the normalized isometry on the operator system, carries each labeled vector to its primed vector. This records an identity of actual specialized endomorphisms, in addition to the de Rham equation. No restriction on the Newton polygon has been used.

For the local sums retain the selected embeddings of M with their absolute weights. If a later algebraic operation introduces another coefficient field F' , work over MF' and take all extensions of those selected embeddings. The common equation, the equality $|R|_v = 1$, and the local norm bounds are preserved by restriction of absolute values, and the new weights sum to the old ones. This step requires no new choice of a Frobenius lift. Thus the field indexing the selected embeddings can be large, while the field defining the algebraic tuple has polynomial degree. The degree of the normal closure never enters the coordinate-degree, height, or grouping estimates.

There are in total polynomially many groups. Apply [Lemma 4.5](#) and select one to obtain (4.9). Put the ordinary coordinates z, P and, where present, z', P', R in U . The constants B, B' belong to fixed data; T is a coefficient of the equation, not an extra variable. Equations (4.12) require the base coordinates to be those of the selected inverse branches. Their inclusion keeps the number of variables and equations fixed and prevents another sheet of the rational parameter from being treated as the chosen germ. Equation (4.13) follows from [Proposition 4.3](#), its conjugate version, and the local and global bounds for R .

Coefficient heights. It remains to bound the common series, uniformly across its selected places. Outside the fixed exceptional set, the algebraic chart series and the connection matrix in that chart have integral Taylor coefficients. If $Z(X) = \sum Z_j X^j$ is an identity-normalized horizontal matrix with integral ordinary connection matrix, its recursion shows inductively that

$$Z_j \in (j!)^{-1} M_r(\mathcal{O}_v).$$

For example $(j+1)Z_{j+1}$ is a sum of Z_k times integral connection coefficients for $k \leq j$, and $j!/k!$ is integral. The same holds for inverse horizontal matrices and the fixed rank-five

system. Hence $\|Z_j\|_v \leq \exp(j\alpha_p)$ at a good prime. Integral algebraic chart series satisfy the stronger bound $\|z_j\|_v \leq 1$.

At every fixed exceptional finite place the implicit function theorem gives a positive radius of convergence for each algebraic chart series; its coefficients and those of the connection are bounded by c_v^{j+1} for a fixed $c_v \geq 1$. The same recursion then bounds a horizontal coefficient by $c_v^j |1/j!|_v$, after increasing c_v . At infinity the ordinary analytic solution and the chart series have fixed positive convergence radii, so their coefficients grow at most exponentially. Since only finitely many c_v differ from one, summing these bounds and using $\sum_p v_p(j!) \log p = \log(j!)$ gives, for example,

$$h(Z_0, \dots, Z_j, z_0, \dots, z_j) \leq C(j+1) \log(j+2).$$

Only this polynomial estimate for logarithmic heights is needed. In the second series $b = aR$, the coefficient of a^j has an additional factor R^j , explaining the linear degree bound. The remaining coefficients are fixed algebraic coefficients, together with the four rational constants T in the constant case. Their affine heights, including scale, have just been bounded. Polynomial operations of fixed degree and concatenation through order l preserve a polynomial height bound. This proves (4.14) for the complete coefficient tuple.

Local tails. At an unexceptional finite place put $M_v = \max(1, \|U\|_v)$. The ultrametric inequality and $|R|_v = 1$ give, for a fixed integer b and every l ,

$$\left\| \sum_{j \geq l} a^j e_j(U) \right\|_v \leq M_v^b |a|_v^l \exp(l\alpha_p).$$

For $l = 0$ the tail is the entire common relation and vanishes at the target. For $l \geq 1$ the constant matrix T does not occur in the tail. The displayed inequality follows by taking the maximum over $j \geq l$; (4.6) makes those bounds decrease. There is no multiplicative constant greater than one at every prime. Fixed products or tensor expressions satisfy the same estimate after increasing b and the fixed coefficient bounds, because the total Taylor index controls the factorial bound.

At the finitely many exceptional places and at infinity, choose the disks strictly inside a fixed convergence disk. Cauchy estimates, or the local differential recursion, then give the same tail estimate with α_p replaced by a fixed constant; at infinity the geometric-series sum contributes one fixed further constant. Consequently one may take

$$b_v \leq C \log M_v + C\alpha_p$$

at the unexceptional primes, with fixed additional constants at the other places. The contribution of α_p over all selected primes is $O(1 + \log^2(dh))$ by (4.8) and the calculation in Lemma 4.5. Summing and using (4.13) proves (4.15)–(4.16). All operations used involve a fixed number of fixed ordinary systems and their fixed tensor constructions, as asserted. $\square$

The equations in Proposition 4.6 provide common arithmetic relations, including at primes where the reduction is supersingular. Their nonidentity on the algebraic varieties encountered by interpolation is a separate requirement, addressed in the next section.

5 Auxiliary markings and the height bound

We prove the nonidentity required by Theorem 3.1 for the common relations of Proposition 4.6, and then deduce the height bound. Single and product monodromy handle the constant relations and the paired relations away from the finitely many dominating isogeny correspondences. The remaining case requires one auxiliary level.

Here is the role of that level. At a common reduction, Rosati positivity makes the rational span of the centered and target endomorphism planes nondegenerate. We arrange that the sum of their $\mathbb{Z}_\ell$ -lattices is nevertheless saturated of rank four, with degenerate reduction modulo a fixed prime ℓ . A functional identity would force the composite of relative Frobenius with the inverse of a multiplier- m isogeny to act by conjugation as one common sign on the rational span. The positive sign forces a rational scalar, incompatible with its multiplier p/m , where p is the residue characteristic and m is a fixed correspondence multiplier prime to p . The negative sign makes the rank-four lattice an orthogonal eigensummand of a unimodular lattice, whose reduction modulo ℓ must be nondegenerate. Thus either sign gives a contradiction. The marking imposes no restriction on the quaternion algebra of the target.

5.1 A finite-field incidence

The required marking comes from the following incidence statement. Its acting element must lift to the spin group because symplectic level markings act on the quadratic system through $\mathrm{Sp}_4 \longrightarrow \mathrm{SO}_5$.

Lemma 5.1. *Let k be a finite field of odd characteristic, let (V, q) be a nondegenerate quadratic space of dimension five, let $A \subset V$ be a nondegenerate 2-plane, and let $T \subset V$ be any 2-plane. There is an element*

$$u \in \mathrm{im}(\mathrm{Spin}(V)(k) \longrightarrow \mathrm{SO}(V)(k))$$

such that $A \cap uT = 0$ and $A + uT$ is a degenerate 4-plane, with radical of dimension one. The conclusion includes restricted quadratic forms on T of ranks zero, one, and two.

Proof. Write $b(v, w) = q(v + w) - q(v) - q(w)$. The nondegenerate ternary space $A^\perp$ is isotropic over k , so it has a decomposition

$$A^\perp = \langle e, f \rangle \perp \langle c \rangle, \quad q(e) = q(f) = 0, \quad b(e, f) = 1, \quad q(c) = \delta \neq 0.$$

Here and below angle brackets around vectors mean their linear span. For any $v, w \in A$, the plane

$$T_* = \langle e + v, c + w \rangle$$

is disjoint from A , and $A + T_* = e^\perp$. This hyperplane has radical ke . We can arrange that T_* and T have isometric restricted forms. A nondegenerate binary form over k represents every nonzero scalar: in the split case use the form xy , and in the anisotropic case use surjectivity of the norm $k_2^\times \rightarrow k^\times$. The value zero is represented by the zero vector. Consequently the following choices are possible:

- (i) If $q|_T = 0$, take $v = 0$ and $q(w) = -\delta$.
- (ii) If $q|_T$ has rank one, write it as $\mathrm{diag}(0, \beta)$ with $\beta \neq 0$, take $v = 0$, and choose $q(w) = \beta - \delta$.
- (iii) If $q|_T$ has rank two, choose a vector of norm δ in T and an orthogonal complement of norm $\alpha \neq 0$. Take $q(v) = \alpha$ and $w = 0$. Then $q|_{T_*} = \mathrm{diag}(\alpha, \delta)$.

Witt extension, which applies also to isometries between degenerate subspaces of a nondegenerate space in characteristic different from two [EKM08, Theorem 8.3], extends the chosen isometry $T \rightarrow T_*$ to an element of $\mathrm{O}(V)(k)$. If its determinant is -1 , compose it with reflection in an anisotropic vector of $T_*^\perp$. Such a vector exists: a three-dimensional subspace of a nondegenerate five-dimensional space cannot be totally isotropic. The reflection fixes T_* pointwise. We have therefore obtained $u_0 \in \mathrm{SO}(V)(k)$ with $u_0 T = T_*$.

It remains to correct the spinor norm. The setwise stabilizer of every 2-plane E in $\mathrm{SO}(V)(k)$ has surjective spinor norm. To see this, choose a hyperbolic plane $\langle r, s \rangle$, normalized by $b(r, s) = 1$, and consider

$$h_z(r) = zr, \quad h_z(s) = z^{-1}s, \quad h_z|_{\langle r, s \rangle^\perp} = 1 \quad (z \in k^\times).$$

If R_v denotes reflection in v , then $h_z = R_{r+s}R_{r+zs}$, so its spinor norm is z modulo squares. When E is nondegenerate, take this hyperbolic plane inside the nondegenerate ternary space $E^\perp$; then h_z fixes E pointwise. When E is degenerate, write $E = \langle r, a \rangle$ with $0 \neq r \in \mathrm{rad}(E)$. Choose s_0 with $b(r, s_0) = 1$ and $b(a, s_0) = 0$, and put $s = s_0 - q(s_0)r$. Then $q(s) = 0$ and $a \perp \langle r, s \rangle$, so h_z preserves E setwise. This also treats the totally isotropic case.

Apply the preceding construction to $E = T_*$, choosing the spinor norm of h_z to be the inverse of that of u_0 . The element $u = h_z u_0$ still maps T to T_* and has trivial spinor norm. To see the lifting assertion directly, write an orthogonal transformation of determinant one as an even product of reflections [SP20, Theorem 1.32]. The corresponding product in the Clifford algebra has Clifford norm equal to the product of their quadratic norms. If this product is a square c^2 , rescaling the Clifford element by c^{-1} gives a norm-one element, hence an element of $\mathrm{Spin}(V)(k)$ with the same orthogonal action [SP20, Corollary 9.6 and Definition 9.7]. Conversely a spin lift has trivial spinor norm. Thus u has the required lift. In dimension five over k , this spin group is the split group Sp_4 . $\square$

5.2 Fixing the level and the exceptional primes

We first specify the order of the fixed choices. Apply Lemma 2.1 to obtain a smooth connected fine-level curve S_* and its family. If the quaternionic-division locus is finite, both the finiteness assertion and the height bound below hold after choosing constants. We therefore assume in the construction that the locus is infinite, and choose $t_{0,*}$ away from the finite omissions. Fix its labeled endomorphism pair and all the conjugates of these data. Apply Proposition 2.4 to the finite list of underlying image curves. Choose integral representatives of the resulting rational similitudes and let $\mathcal{M}$ be their finite set of positive polarized multipliers. This step precedes the auxiliary level: passing to that level does not enlarge the list of multipliers required for the underlying image curves.

For a rational endomorphism plane E_t , let

$$L_t = E_t \cap \Lambda_t$$

be its saturated lattice in the integral quadratic local system of Lemma 2.2. The integral lattice is understood with the fixed finite set of bad denominator primes removed. Saturation implies that $L_t/\ell L_t$ is a 2-plane in $\Lambda_t/\ell \Lambda_t$ at every remaining prime ℓ . This holds even when its quadratic form becomes degenerate. The small generating pair from Proposition 4.2 need not be a basis of L_t and will not be used to define its reduction.

Proposition 5.2. *There is a fixed odd prime ℓ and a fixed geometrically connected finite étale cover $S \rightarrow S_*$ parameterizing symplectic markings at level ℓ , with a fixed lift t_0 of $t_{0,*}$, such that every quaternionic-division point of S_* outside a finite set has a lift $t \in S$ satisfying the following condition. In the common quadratic space supplied by the markings,*

$$\dim(\overline{L}_{t_0} + \overline{L}_t) = 4, \quad q|_{\overline{L}_{t_0} + \overline{L}_t} \text{ is degenerate.} \quad (5.1)$$

The prime ℓ is prime to every $m \in \mathcal{M}$. At every prime $p \neq \ell$ of good reduction for these fixed models where t and t_0 have the same marked reduction, the corresponding two saturated $\mathbb{Z}_\ell$ -lattices have sum M satisfying

$$M = \Lambda_\ell \cap (M \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell), \quad \mathrm{rank} M = 4, \quad M/\ell M \text{ is degenerate.} \quad (5.2)$$

These assertions are preserved by simultaneous field conjugation.

Proof. The geometric monodromy subgroup in $\mathrm{Sp}_4(\mathbb{Z})$ is finitely generated, because S_*^{an} has finitely generated fundamental group, and is Zariski dense by [Proposition 2.3](#). Strong approximation for a finitely generated Zariski-dense subgroup of a simply connected absolutely almost simple group gives surjectivity onto $\mathrm{Sp}_4(\mathbb{F}_\ell)$ for all sufficiently large primes ℓ [[MVW84](#), Theorem, p. 515]. Choose such an ℓ , avoiding 2, the discriminant of the ambient quadratic lattice, the discriminant of $L_{t_0,*}$, and every multiplier in $\mathcal{M}$. After a fixed extension containing the relevant root of unity, full symplectic markings with the prescribed pairing form a torsor under $\mathrm{Sp}_4(\mathbb{F}_\ell)$. Surjective geometric monodromy says exactly that its pullback to S_* is geometrically connected.

Fix a marking above $t_{0,*}$. Its plane is nondegenerate modulo ℓ . For any target, its saturated plane still has dimension two modulo ℓ , and [Lemma 5.1](#) supplies a change of marking in the image of $\mathrm{Sp}_4(\mathbb{F}_\ell)$ giving (5.1). All markings occur on the same cover S . Deleting finitely many points to choose frames and affine charts excludes only finitely many points downstairs. In particular, ℓ has not been chosen to avoid the discriminants of the varying target planes.

At a common marked reduction, smooth proper base change identifies each integral ℓ -adic cohomology lattice with that of the common reduction. The mod- ℓ identifications agree with the markings. Take $\mathbb{Z}_\ell$ -bases of the two saturated planes. Their four reductions are independent by (5.1), so some 4-row minor of their coordinate matrix is an ℓ -adic unit. These four vectors extend to a basis of Λ_ℓ ; their sum M is a direct summand, and therefore satisfies the saturation equality in (5.2). Its reduction is precisely the degenerate space in (5.1). This proves local saturation at ℓ ; no global saturation assertion about the sum of integral lattices is needed. Simultaneous conjugation carries both markings, both lattices, and their common pairing to the conjugate data, preserving independence and degeneracy. $\square$

We have now fixed the incidence condition and its integral consequence. Enlarge the fixed field K for the cover, the chosen lift t_0 , and its labeled pair, and use the notation of [Section 4](#) on S . Before grouping proximity places, we make one further finite exclusion: every isogeny of an allowed multiplier between the reduced centers must lift to one of finitely many characteristic-zero isogenies between them. The next lemma provides this assertion for the maps themselves.

Lemma 5.3. *Fix finitely many pairs of principally polarized abelian surfaces over a number field and finitely many positive integers m . Outside a fixed finite set of rational primes, every polarized isogeny of multiplier m between the geometric reductions of a fixed pair is the specialization of a characteristic-zero polarized isogeny of that multiplier between the pair. The set of these characteristic-zero maps is finite. The assertion concerns the maps themselves, and holds for supersingular reductions as well.*

Proof. It suffices to treat one pair (A_0, λ_0) , (A'_0, λ'_0) and one m . Choose models over a localization of the ring of integers, with good reduction and with m invertible. A polarized isogeny φ with $\varphi^* \lambda'_0 = m \lambda_0$ has degree m^2 and kernel contained in $A_0[m]$. After a fixed finite extension and localization, all subgroup schemes of this fixed order and the required isotropic type in $A_0[m]$ are the spreads of the finitely many characteristic-zero ones. Indeed $A_0[m]$ is finite étale, and after splitting it these are subgroups of one fixed finite group.

Each possible kernel gives a fixed polarized quotient (D, λ_D) , with the quotient map pulling λ_D back to $m \lambda_0$. The remaining data are polarized isomorphisms $(D, \lambda_D) \simeq (A'_0, \lambda'_0)$. Here isomorphisms are group isomorphisms of abelian schemes, preserving the zero section and the polarizations. Their scheme is unramified by rigidity and is of finite type: for relatively ample line bundles representing the polarizations, the restriction of their product to such a graph is numerically twice the source line bundle, and hence has a fixed Hilbert

polynomial. Thus these graphs lie in one relative Hilbert scheme. The fibers of the polarized-isomorphism scheme are torsors under the finite polarized group-automorphism groups [Mil86, Proposition 17.5], so the scheme is quasi-finite.

A quasi-finite scheme of finite type over a ring of integers, with finite generic fiber, becomes the spread of that generic fiber after deleting finitely many primes. For completeness, discard its finitely many vertical irreducible components; its finitely many generic points then spread to a finite scheme after localization, and deleting the support of the remaining complement gives the assertion. Further localization makes this scheme finite étale. After a fixed extension splitting its generic fiber, this finite étale scheme is a disjoint union of sections: each component has generic degree one over the normal base. Its geometric points are therefore exactly the specializations of the fixed characteristic-zero maps, including every polarized-automorphism choice. Applying this to the finite list of kernels proves the lemma. $\square$

Include the primes excluded by Lemma 5.3, those dividing the elements of $\mathcal{M}$, and ℓ in the fixed exceptional set of Proposition 4.6. Here the center pairs are the finitely many pairs of conjugates of t_0 . The possible center isogenies are now fixed algebraic data. Take the full finite list of these maps between every ordered pair of absolute conjugates of the center; the list is stable under simultaneous absolute Galois conjugation. Enlarging K once more defines all of them. This enlargement adds no geometric center to the list, since the absolute conjugates were included already. The local Frobenius lift in Proposition 4.6 need not fix K : it permutes the conjugate centers and maps in this fixed list. At an embedding ρ of the labeled coefficient field, a selected specialized center map is $\rho(\varphi_0)$ for a member φ_0 of that list. Subdividing by this member makes the algebraic choice common to the selected embeddings. Upon any further coefficient-field extension we retain all extensions of each selected embedding, so the map and its specialization remain compatible and the total weights are unchanged. A chosen multiplier- m isogeny between target fibers also requires only a bounded relative field extension. Its kernel is one of a uniformly bounded number of subgroups of the fixed-size group $A_t[m]$, and the polarized identifications form a torsor under a finite automorphism group of bounded order in dimension two. For the latter bound, such a group acts faithfully on the rank-four integral Betti lattice, and finite subgroups of $\mathrm{GL}_4(\mathbb{Z})$ have bounded order. Thus these choices preserve all polynomial degree bounds in Section 4.

5.3 Nonidentity on every branch

We use three lists of equations. Constant relations will be excluded by single monodromy, and paired relations outside the finite correspondence locus by product monodromy. On that locus we first convert the paired relation to one involving a single transport matrix; the marking will then exclude an identity.

The notation $B = (B_1, B_2)$ and $P = (P_1, P_2)$ means the labeled centered and target pairs, and $Y(a)$ transports back to the center. We retain all four entries of each matrix equation: in the exceptional case they will force the same sign on both entire planes. In every case include the algebraic chart equations from Proposition 4.6: if z denotes affine coordinates on S and $z = \zeta(a)$ is the actual inverse germ of the chosen parameter, include $z - \zeta(a) = 0$. For a second argument include $z' - \zeta'(aR) = 0$. The ambient algebraic conditions put z, z' on the respective curves and impose $a = x(z)$ and $aR = x'(z')$ when appropriate. Only fixed denominators needed for these affine charts are cleared.

The first list, used at archimedean or fixed exceptional places, is

$$\langle B, Y(a)P \rangle = c, \tag{5.3}$$

where the 2-by-2 rational matrix c is common to the chosen group. The second list is

$$\langle B, Y(a)P \rangle = \langle B', Y'(aR)P' \rangle. \quad (5.4)$$

It is used when the target pair of image points is outside all the finitely many dominating isogeny correspondences of [Proposition 2.4](#).

For the third list the target pair belongs to one of those correspondences. We transport its primed vectors to the unprimed fiber using an isogeny, and do the same at the centers. To compare these two operations, their isogenies must have the same specialization. Choose a polarized isogeny φ_s of multiplier $m \in \mathcal{M}$ between the targets, and write I_s for its normalized isometry on the quadratic systems. First adjoin its field of definition to the paired target field and extend every already selected embedding by all its extensions. This costs only a bounded relative degree and preserves every weighted local sum. The previously chosen Frobenius identities restrict to the original labeled tuples, so remain true under these extensions without a new choice of Frobenius lift. At a place of this expanded group the unprimed center and target reduce to the same marked point r of the good fine model, and hence to the same universal fiber A_r . The chosen local automorphism has residue action Frob_p , so the primed center and target both reduce to $r^{(p)}$ and to $A_r^{(p)}$. These are the conjugate identifications of the same source fiber, not independently chosen isomorphisms. The target isogeny extends between the good abelian models; its specialization is an actual multiplier- m map $A_r \rightarrow A_r^{(p)}$. By [Lemma 5.3](#) this map is the specialization of one fixed center isogeny φ_0 , with normalized isometry I_0 . Only now subdivide the expanded embedding set by this finite choice from the conjugate-stable list of actual maps, including each map's polarized quotient identification.

The center and target comparison maps now identify both isogenies with the same specialized map. Functoriality in [Proposition 4.4](#) therefore gives, at this target,

$$I_0 Y(a) = Y'(aR) I_s. \quad (5.5)$$

Put $D_i = I_0^{-1} B'_i$ and $\hat{P}_j = I_s^{-1} P'_j$ at the target. The third list is the single-argument system

$$\langle B, Y(a)P \rangle = \langle D, Y(a)\hat{P} \rangle. \quad (5.6)$$

In this system D is fixed and $\hat{P}$ is an additional pair of vector coordinates in the unprimed frame; the second argument can be dropped. Its target value is the realization of actual rational endomorphisms, with denominators bounded in terms of m and with the same Rosati norm bounds as P' . More explicitly, if $\hat{\alpha} = \varphi_s^{-1} \alpha' \varphi_s$, then $m\hat{\alpha} = \varphi_s^\dagger \alpha' \varphi_s$ is integral and $q_{\lambda_s}(\hat{\alpha}) = q_{\lambda'_s}(\alpha')$. Thus [Propositions 4.3](#) and [4.6](#) apply to this new tuple. No isogeny is assumed to exist along a containing algebraic variety used in interpolation.

Proposition 5.4. *For a selected marked target as in [Proposition 5.2](#), each applicable list above, formed from a nonempty selected group of places, satisfies the all-branch nonidentity hypothesis of [Theorem 3.1](#). More precisely, let W be any irreducible algebraic subvariety of the indicated affine ambient conditions which contains the target and on which a is nonconstant. On every branch over the generic point of every component of $W \cap \{a = 0\}$, at least one equation in the applicable list, including the chart equations, is not an identity.*

Proof. *From a branch identity to monodromy.* Suppose that every equation is an identity on such a branch. On the normalization near a general point of its divisor, a vanishes and all affine coordinates, including R when present, are regular. The series in a and aR therefore converge after a complex embedding in a neighborhood of that point. In a transverse parameter on the normalization, the formal identity makes every coefficient zero at the generic point of the divisor. Convergence therefore gives an identity on a nonempty analytic open.

Any fixed denominator cleared in a chart is nonzero at the target, hence is a nonzero function on W and has finite order on each normalization branch. It cannot turn a nonzero series into an identity. If one of the chart equations is not an identity we already have the desired conclusion; otherwise they identify the series arguments with the actual projections to S and S' . The first projection is nonconstant since a is nonconstant.

Pass to a smooth dense open of W on which these projections and the algebraic frames are defined. An irreducible complex algebraic variety has connected smooth locus, so the analytic identity continues on its universal cover. At a fixed point of this open the vector coordinates are single-valued, whereas continuation changes the transport matrices by the monodromy of the pulled-back systems. A dominant map to a curve has fundamental-group image of finite index after deleting suitable closed subsets and passing to a resolution; thus [Proposition 2.3](#) applies on this open. The vector pairs are generically independent, because they are independent at the target. A pair minor nonzero at the target is not identically zero on W , so its nonvanishing locus meets this analytic open even if the pair vanishes on the divisor itself. They need not remain independent at every boundary point of W .

Constant relations and independent projections. For (5.3), fix a general point of this open. The orbit of the transport matrix under continuation has Zariski closure a translate of SO_5 . For nonzero vectors b, v in its standard representation, the function $g \mapsto \langle b, gv \rangle$ is not constant. Indeed, the span of the differences $gv - v$ is invariant, and is the whole representation by irreducibility and the absence of invariant vectors. A constant matrix coefficient would annihilate this span. Applying this to B_1 and P_1 contradicts the continued identity.

For (5.4), if the second projection is constant, only the first system varies and the same argument applies. If it is nonconstant, [Proposition 2.3](#) gives product monodromy unless the closure of the paired image is contained in a dominating isogeny correspondence. In the latter case [Proposition 2.4](#) puts the image in the fixed finite list; since these correspondences are closed, the target also belongs to that list, contrary to this case's defining assumption. For product monodromy, fix the second transport matrix and vary the first. The resulting constant matrix coefficient is again impossible.

The exceptional relation forces a common sign. Consider now (5.6). At a general point z of W , write $Y_z : \mathcal{V}_z \rightarrow \mathcal{V}_{t_0}$ for transport. The continued identity holds with transport gY_z for every $g \in \mathrm{SO}(\mathcal{V}_{t_0})$. The standard representation is absolutely irreducible, so Burnside's theorem says that these matrices span $\mathrm{Hom}(\mathcal{V}_z, \mathcal{V}_{t_0})$. For each i, j , the linear functional

$$M \mapsto \langle B_i, MP_j \rangle - \langle D_i, M\hat{P}_j \rangle$$

therefore vanishes on every linear map M between these two spaces. The corresponding rank-one tensors are equal. Identifying their first factors with covectors by the fixed center trace pairing, we obtain

$$B_i \otimes P_j = D_i \otimes \hat{P}_j \quad (1 \leq i, j \leq 2). \quad (5.7)$$

These are polynomial equations in the algebraic vector coordinates. They hold on a dense open of W , and therefore at the target itself. There, all four pairs of factors are nonzero. The (1, 1) equation gives $D_1 = c_0 B_1$ and $\hat{P}_1 = c_0^{-1} P_1$ for some nonzero scalar c_0 ; the (2, 1) and (1, 2) equations give the same scalar for the other indices. The self-pairing of B_1 is a nonzero rational number, and is unchanged by conjugation or the isometry I_0 . Thus $c_0^2 = 1$, and

$$D_i = \varepsilon B_i, \quad \hat{P}_j = \varepsilon P_j, \quad \varepsilon \in \{1, -1\}. \quad (5.8)$$

The (2, 1) and (1, 2) equations are essential here: they couple the two labels and give a single sign for both pairs.

Specialization to a Frobenius quasi-isogeny. We now use that the vectors in (5.8) come from endomorphisms. De Rham realization on the rational endomorphism algebra of a characteristic-zero abelian variety is faithful: base change to $\mathbb{C}$, compare with Betti cohomology, and use the faithful action on its rational first homology. Conjugation by an isogeny preserves the Rosati self-adjoint subspace. Consequently (5.8) asserts equalities of actual rational endomorphisms in the center and target fibers. These equalities specialize under good reduction.

Choose any place in the nonempty selected group and let p be its residue characteristic. The unprimed center and target have a common marked reduction $\overline{A}$, and the primed reductions identify with its p -twist. Write $F : \overline{A} \rightarrow \overline{A}^{(p)}$ for relative Frobenius, and write $\overline{\varphi}_0$ for the common specialization of φ_0 and φ_s . The map

$$u = \overline{\varphi}_0^{-1} F \in \text{End}^0(\overline{A})$$

is a self-quasi-isogeny with polarized multiplier p/m . Relative Frobenius transports the unprimed labeled endomorphisms to their primed specializations. The specialized equalities (5.8) therefore say that the conjugation action g of u on the quadratic $\mathbb{Q}_\ell$ -space acts by ε on both specialized endomorphism planes. The sign action extends to their saturated $\mathbb{Z}_\ell$ -lattices; the chosen small pairs need not be bases of those lattices. By Lemma 2.2, g belongs to SO_5 .

Excluding the two signs. Let T be the rational span of those two planes in $\text{End}^0(\overline{A})$. Its ℓ -adic realization has dimension four by Proposition 5.2. The trace form on its rational symmetric endomorphisms is positive definite by Rosati positivity in arbitrary characteristic [Mil86, Theorem 17.3]; hence T is nondegenerate over $\mathbb{Q}$, and its realization T_ℓ is nondegenerate over $\mathbb{Q}_\ell$. The one-dimensional perpendicular is consequently g -stable. Since $\det g = 1$ and $\det(g|_{T_\ell}) = \varepsilon^4 = 1$, g acts as $+1$ on that perpendicular.

If $\varepsilon = +1$, it follows that $g = 1$. The kernel of the conjugation representation of GSp_4 on this quadratic space is its scalar center by Lemma 2.2. Put $\mu = p/m$. With the covariant convention of Section 2, the action on first cohomology is

$$T_u = \mu(u^*)^{-1} = (u^\dagger)^*;$$

the equality follows from $u^\dagger u = [\mu]$. Thus $(u^\dagger)^* = \text{cid}$ for some $c \in \mathbb{Q}_\ell$. The ordinary cohomological trace of the rational quasi-endomorphism $u^\dagger$ is rational [Mil86, Proposition 12.9], so $4c = \text{Tr}(u^\dagger) \in \mathbb{Q}$. Faithfulness gives $u^\dagger = \text{cid}$, and applying Rosati gives $u = \text{cid}$. Its multiplier would satisfy $c^2 = p/m$. This is impossible because $p \nmid m$, so the p -adic valuation of p/m is odd.

If $\varepsilon = -1$, g is an involution with negative eigenspace exactly T_ℓ . Both F and $\overline{\varphi}_0$ have multipliers prime to ℓ , so their actions are invertible on the integral Tate modules; conjugation by u preserves the integral quadratic lattice Λ_ℓ . This lattice is unimodular by the choice of ℓ . Since ℓ is odd, the integral projectors $(1+g)/2$ and $(1-g)/2$ give an orthogonal direct sum

$$\Lambda_\ell = \Lambda_+ \perp \Lambda_-, \quad \Lambda_- = \Lambda_\ell \cap T_\ell.$$

Each summand is unimodular, so $\Lambda_-/\ell\Lambda_-$ is nondegenerate. But Proposition 5.2 identifies Λ_- with the sum M of the two marked saturated plane lattices, whose reduction is degenerate by (5.2). This is the required contradiction.

Both cases contradict the assumed identity. The use of Rosati positivity and rational endomorphism trace is valid for every reduction type; no classification or ordinarity assertion about Frobenius has been used. $\square$

5.4 Application of interpolation and descent

Theorem 5.5. *Fix a reduced irreducible closed Hodge-generic curve $C \subset \mathcal{A}_2$ over $\overline{\mathbb{Q}}$, and let C^ν be its normalization. For a fixed nonnegative ample logarithmic height h on the smooth*

projective completion of C^ν and a fixed field of definition K , there are constants $C_1, C_2 > 0$ such that every point $s \in C^\nu(\overline{\mathbb{Q}})$ whose coarse image has full geometric rational endomorphism algebra an indefinite quaternion division algebra over $\mathbb{Q}$ satisfies

$$\max\{2, h(s)\} \leq C_1 \max\{2, [K(s) : K]\}^{C_2}.$$

The same assertion holds on the fixed fine-level curves used above, and for a fixed projective height of the coarse points. The constants do not depend on the quaternion algebra, its discriminant, or the target endomorphism order.

Proof. If the target locus is finite, enlarge the constants to cover it. Otherwise use the initial family and center, the finite multiplier list, and then the fixed prime and connected marking cover, in the order already specified. For each target choose one lift satisfying Proposition 5.2; this does not enlarge any fixed datum. Remove the finite exceptional targets of the affine charts, and suppose that its height h exceeds a sufficiently large fixed polynomial in $d = \max\{2, [K(t) : K]\}$. The fields generated by the labeled single or paired point tuples in Proposition 4.6 have degree polynomial in d . A normal closure is used only to index selected embeddings. A chosen fixed-multiplier isogeny requires only a bounded relative extension over the field of its two point arguments, so adjoining it preserves this polynomial bound. Selected embeddings are extended by all extensions; the new weights above an old embedding sum to its old weight, preserving the normalized smallness sum.

Arithmetic inputs. Apply Proposition 4.6, with the fixed exceptional primes chosen above. Partition first into its constant and Frobenius groups, and, if needed, subdivide the latter by the finite center-isogeny choice. There are polynomially many groups, and one supplies a common list of equations and a selected weighted set $\mathcal{V}$ with

$$\sum_{v \in \mathcal{V}} w_v \log \frac{1}{|a|_v} \geq \frac{h}{Q_1(d)}, \quad \sum_{v \in \mathcal{V}} w_v \log^+ \|U\|_v \leq Q_2(d)(1 + \log^c h).$$

The tuple (a, U) has global logarithmic height at most $Q_3(d)h$. The coefficient degrees are $O(j+1)$, and the complete affine coefficient tuple through order l , including scale, has logarithmic height at most $Q_4(l, d)(1 + \log^c h)$. The tails satisfy the hypotheses of Theorem 3.1. These assertions also hold for the transformed target pair $\hat{P}$ in the exceptional case: its fixed denominator and preserved Rosati norms give exactly the size hypotheses of Proposition 4.3, and only the original ordinary connection in a is used. In the other paired case $|R|_v = 1$ at every selected place. Thus its two ordinary systems are evaluated at a and aR , as required. The chart equations use fixed algebraic ordinary germs.

Nonidentity and interpolation. Start the interpolation argument inside the fixed affine algebraic conditions on the base coordinates described above. They have bounded complexity. No inverse Gram matrices or inverse minors of the target endomorphism pairs have been adjoined. Every irreducible subvariety encountered in the dimension descent contains the target. If a varies there, Proposition 5.4 supplies nonidentity on every relevant branch, including branches on which a chart equation already fails. If a is constant, the stopping case in Theorem 3.1 applies directly.

Theorem 3.1 now gives $h \leq C_3 d^{C_4}$ for the selected lifts. Its constants are uniform because the cover, centers, allowed ordinary systems, and isogeny multipliers are fixed.

Descent from the selected lifts. The degree of the marking cover is fixed. If t_* is the image of t , then

$$[K(t) : K] \leq (\deg(S \rightarrow S_*)) [K(t_*) : K].$$

Choose the ample height upstairs as the height of the pullback of an ample divisor downstairs; it equals the downstairs height up to a bounded error, and arbitrary fixed ample heights on

these curves are comparable up to fixed positive constants and bounded errors. Thus the bound for the selected lift descends to t_* . The further normalization and fine-level maps descend in the same way by [Lemma 2.1](#). For any other lift $\tilde{t}$ in a fixed cover, height comparison and $[K(t_*) : K] \leq [K(\tilde{t}) : K]$ give the same numerical bound. Thus only the selected lifts require (5.1). At singular coarse points there are finitely many normalization preimages, all included among the finite exceptions. Increasing the constants includes all finitely many omitted points. The absolute height bound for a selected lift also holds for all its conjugates over the fixed data field, even when those conjugates were not selected relative to the original marking of t_0 ; no fresh incidence choice is needed to deduce that numerical bound. All fixed enlargements of the data field K change the degree bounds by at most fixed factors. $\square$

6 From heights to finiteness

We now turn the height estimate into a bound on degrees. The rational objects to be counted are ordered pairs of integral Hodge vectors in the rank-five variation. Their two orthogonality equations cut the period quadric in a conic. A semialgebraic family of such pairs with one real parameter therefore sweeps out an algebraic subset of dimension at most two, whereas full monodromy forces the period image to be Zariski dense in the three-dimensional quadric.

Throughout this section the curve, family, and all auxiliary covers are fixed over a number field K . We choose the ample height on S from a fixed projective embedding defined over K , and write

$$h(t) = \max\{2, h_{\bar{S}}(t)\}, \quad d(t) = \max\{2, [K(t) : K]\}.$$

This choice is invariant under $\text{Gal}(\overline{\mathbb{Q}}/K)$; changing to it preserves the preceding estimates by the height comparisons in [Lemma 2.1](#). Constants in this section depend on the fixed data.

6.1 Flat coordinates at all complex embeddings

The Rosati bound in [Proposition 4.2](#) controls the Hodge norms of our vectors. Counting requires their numerical sizes in integral flat bases. Near a puncture, passing between these two bounds also requires control of the dual Hodge norm at each complex embedding.

Write $b = \langle \cdot, \cdot \rangle$ for the trace pairing of (2.2). In a flat trivialization of $\mathbb{V}_{\mathbb{Z}}$, let $V_{\mathbb{Z}}$ be the corresponding fixed lattice and $V_{\mathbb{C}} = V_{\mathbb{Z}} \otimes \mathbb{C}$. The period quadric of (2.4) is

$$Q_V = \{[w] \in \mathbb{P}(V_{\mathbb{C}}) : b(w, w) = 0\}, \quad \dim_{\mathbb{C}} Q_V = 3.$$

The period line is the line corresponding to $H^{2,0}$, with the Tate twist understood. A real vector of type $(0,0)$ is orthogonal to this line. In particular the two vectors of [Proposition 4.2](#) satisfy these equations, and their Gram matrix for b is positive definite by [Lemma 2.2](#).

Here the counting height is the *multiplicative rational coordinate height*. For $y = (y_1, \dots, y_m) \in \mathbb{Q}^m$, with $y_i = a_i/b_i$ in lowest terms and $b_i > 0$, put

$$H_{\text{rat}}(y) = \max_i \max\{|a_i|, b_i\}.$$

For an integral tuple it is $\max\{1, \|y\|_{\infty}\}$. This height is distinct from the absolute logarithmic algebraic heights used in the arithmetic argument.

Lemma 6.1 (Flat-coordinate bound). *There is a finite collection of simply connected complex charts covering $S(\mathbb{C})$, each equipped with an integral flat basis of $\mathbb{V}_{\mathbb{Z}}$, whose period maps are definable in $\mathbb{R}_{\text{an}, \text{exp}}$. There are constants $c, M > 0$ such that, for every QM point $t \in S(\overline{\mathbb{Q}})$, the ordered pair of integral vectors supplied by [Proposition 4.2](#) has flat coordinate tuple $y \in \mathbb{Z}^{10}$ satisfying*

$$H_{\text{rat}}(y) \leq c(d(t)h(t))^M \tag{6.1}$$

in each of these charts containing t . The same bound holds for their conjugate vectors at every conjugate of t over K .

Proof. We first describe the charts and their metric estimates, including the punctures of S . A compact subset away from the finitely many points of $\bar{S} \setminus S$ has a finite cover by relatively compact simply connected coordinate disks. Shrink these disks inside slightly larger disks on which the flat trivializations and the period maps are holomorphic. Their real and imaginary parts are restricted analytic functions, hence definable.

At a puncture choose a rational function u on $\bar{S}$ which is a local parameter there. Enlarge K once to define these finitely many choices. The local monodromy is quasi-unipotent. On a fixed local cover $u = w^e$ it is unipotent, say $\exp N$, with N nilpotent. The one-variable nilpotent orbit theorem [Sch73, Theorem 4.9] gives, in the compact dual, the expression

$$\Phi(u) = \exp\left(\frac{\log w}{2\pi i} N\right) \Psi(w), \quad (6.2)$$

where Ψ extends holomorphically across $w = 0$. Cover a sufficiently small punctured u -disk by finitely many open angular sectors of width less than 2π , and choose a branch of w and of $\log w$ on each. The exponential in (6.2) is a polynomial in $\log w$. The branch of w is algebraic, its argument is bounded, and $\log |w|$ is definable in $\mathbb{R}_{\text{an,exp}}$. After shrinking the radius, Ψ is restricted analytic in an affine chart about $\Psi(0)$. Thus the projective map (6.2) is definable. These local covers are used only to describe functions on the sectors; no point-dependent global cover is introduced.

Write $\|\cdot\|_{H,u}$ for the Hodge norm. Schmid's abstract one-variable norm estimate [Sch73, Theorem 6.6'], applied both to $\mathbb{V}$ and to its dual, bounds the norms of each fixed flat basis and dual basis by a power of $1 + \log(1/|u|)$, uniformly on each sector. Indeed every fixed flat vector belongs to some step of the monodromy weight filtration, and that theorem supplies such a power; only finitely many basis vectors occur. Passing from w to $u = w^e$ merely changes the constants. Consequently, with $e_1, \dots, e_5$ an integral flat basis and $e_1^*, \dots, e_5^*$ its dual, we have fixed $c_0, r > 0$ such that

$$\max_i \{\|e_i\|_{H,u}, \|e_i^*\|_{H,u}^\vee\} \leq c_0(1 + \log(1/|u|))^r. \quad (6.3)$$

On the compact charts the corresponding bound is constant. In particular (6.3) controls the metric and its inverse in flat coordinates.

For completeness, the logarithm in this bound can be controlled at an individual complex embedding, not merely on average. Let $F = K(t)$ and let $\sigma : F \hookrightarrow \mathbb{C}$ fix the chosen embedding of K . Whenever $\sigma(t)$ belongs to the sector, $u(t) \neq 0$, and the absolute height identity $h(u(t)^{-1}) = h(u(t))$ gives

$$\begin{aligned} \log^+ \frac{1}{|\sigma(u(t))|} &\leq [F : \mathbb{Q}] h(u(t)) \\ &\leq c_1 d(t) h(t). \end{aligned} \quad (6.4)$$

The first inequality follows by retaining that archimedean summand in the height of $u(t)^{-1}$, using the usual nonsquared absolute value. For the second, the fixed rational function u defines a morphism $\bar{S} \rightarrow \mathbb{P}^1$, whose height is bounded above by a constant times $1 + h_{\bar{S}}$. This also explains why algebraic local parameters were chosen.

Let v be either of the two chosen endomorphism vectors, at any such embedding. If v realizes the symmetric endomorphism α , the Weil operator fixes its type $(0,0)$, and its rationality gives

$$\|v\|_H^2 = b(v, v) = \frac{1}{4} \text{Tr}(\alpha^2) = \frac{1}{4} q_{\lambda_t}(\alpha).$$

The bound in [Proposition 4.2](#) therefore gives $\|v\|_H \leq c_2(d(t)h(t))^{r_2}$. If $v = \sum_i n_i e_i$, the dual norm inequality is

$$|n_i| = |e_i^*(v)| \leq \|e_i^*\|_H^\vee \|v\|_H.$$

Combining this with (6.3) and (6.4) proves a polynomial bound on the *numerical sizes* of all ten integers. This is precisely (6.1).

The vectors are realizations of actual integral endomorphisms. Conjugation preserves integrality, independence, and the rational Rosati trace norms. Although labeling the endomorphisms may require an extension of $K(t)$ of bounded relative degree, each embedding of $K(t)$ extends to that field. Conjugating the labeled pair along any such extension supplies the asserted pair at the conjugate point. Neither the degree estimate for $u(t)$ nor the number of conjugate points is diminished by this labeling extension. $\square$

6.2 Counting distinct curve arguments

We use the following precise specialization of the counting theorem. The structure $\mathbb{R}_{\text{an}, \text{exp}}$ containing our period charts is o-minimal by [vdDMM94, Corollary 5.13]. Let $Z \subset \mathbb{R}^m \times \mathbb{R}^n$ be definable in an o-minimal expansion of the real field, and write π_1, π_2 for its two projections. For every $\eta > 0$ there is $c_Z(\eta)$ such that, if $T \geq 1$ and

$$\Sigma \subset \{(y, z) \in Z : y \in \mathbb{Q}^m, H_{\text{rat}}(y) \leq T\}, \quad \#\pi_2(\Sigma) > c_Z(\eta)T^\eta,$$

then there is a continuous definable path $\beta : [0, 1] \rightarrow Z$ for which $\pi_1\beta$ is semialgebraic and real analytic on $(0, 1)$, while $\pi_2\beta$ is nonconstant. This is [HP16, Corollary 7.2, with $k = 1$ and no family parameter]. The corollary counts distinct second projections and imposes no finite-fiber hypothesis on $Z \rightarrow \mathbb{R}^m$. A constant first projection is allowed, and causes no difficulty below.

Proposition 6.2. *For fixed $A, \kappa > 0$, the set of QM points $t \in S(\overline{\mathbb{Q}})$ satisfying*

$$h(t) \leq Ad(t)^\kappa \tag{6.5}$$

is finite.

Proof. The set in question is stable under $\text{Gal}(\overline{\mathbb{Q}}/K)$. If its degrees were unbounded, take a point t of sufficiently large degree $d = [K(t) : K]$. Its full orbit has d distinct points on $S(\mathbb{C})$, all satisfying (6.5). Assign each conjugate to one chart of [Lemma 6.1](#). If there are J charts, one chart contains at least d/J of these distinct points. For each of them choose a conjugate of the labeled pair at t . Their integral coordinate tuples have

$$H_{\text{rat}}(y) \leq T = c_3 d^B \tag{6.6}$$

for fixed $c_3, B > 0$.

In the selected chart write $z \in D \subset \mathbb{R}^2$ for the curve coordinate and $\Phi(z) \in Q_V$ for the period line. Identify the flat real space with $\mathbb{R}^5$. Consider the fixed definable set

$$Z = \left\{ ((v_1, v_2), z) \in (\mathbb{R}^5)^2 \times D : \begin{array}{l} (b(v_i, v_j))_{i,j=1}^2 \text{ is positive definite,} \\ b(v_1, \Phi(z)) = b(v_2, \Phi(z)) = 0 \end{array} \right\}. \tag{6.7}$$

Orthogonality to a projective line means orthogonality to any nonzero representative; it is a well-defined algebraic incidence condition. The coordinate z is injective on the chart. The tuples just chosen thus form a set $\Sigma \subset Z$ with at least d/J distinct second projections and rational first coordinates bounded as in (6.6).

Choose $\eta > 0$ with $B\eta < 1$. The finitely many chart incidences have a common counting constant, obtained by taking their maximum. For large enough d we have

$$d/J > c_Z(\eta)(c_3 d^B)^\eta.$$

The quoted corollary therefore gives a path $\beta(s) = (\gamma(s), z(s))$ in Z , with γ semialgebraic and z nonconstant. It remains to show that this path would force the period image into a proper algebraic subset of Q_V . The first projection supplies at most one algebraic parameter, and for each nondegenerate pair the orthogonal period lines form a conic.

Let $\mathcal{A} \subset (\mathbb{C}^5)^2$ be the complex Zariski closure of $\gamma((0, 1))$. A semialgebraic set of real dimension at most one has real Zariski closure of dimension at most one, and complexification preserves that dimension. Hence $\dim_{\mathbb{C}} \mathcal{A} \leq 1$. Equivalently, after a finite semialgebraic decomposition, the arc has Nash coordinate functions, which are algebraic over its real parameter; their algebraic closure has transcendence degree at most one.

Crucially, impose nondegeneracy before closing the orthogonality incidence. Let

$$\mathcal{U} = \{(v_1, v_2) : \det(b(v_i, v_j))_{i,j=1}^2 \neq 0\} \subset (\mathbb{C}^5)^2,$$

and define the locally closed algebraic set

$$\mathcal{I} = \{(v_1, v_2, [w]) \in (\mathcal{A} \cap \mathcal{U}) \times Q_V : b(v_1, w) = b(v_2, w) = 0\}.$$

For each $(v_1, v_2) \in \mathcal{A} \cap \mathcal{U}$, their span is a nondegenerate two-space. Its perpendicular is a nondegenerate three-space, whose isotropic lines form a smooth projective conic. Every fiber of $\mathcal{I} \rightarrow \mathcal{A} \cap \mathcal{U}$ therefore has dimension one, so

$$\dim_{\mathbb{C}} \mathcal{I} \leq 2.$$

Let $X \subset Q_V$ be the Zariski closure of its image under the second projection. The dimension of an image and its closure cannot exceed the dimension of its source, whence

$$\dim_{\mathbb{C}} X \leq 2. \tag{6.8}$$

This construction closes only the incidence over $\mathcal{A} \cap \mathcal{U}$; it does not add arbitrary fibers over degenerate pairs. Since the pairs on γ have positive definite Gram matrix, they lie in $\mathcal{U}$, and consequently $\Phi(z(s)) \in X$ for all $s \in (0, 1)$.

The continuous nonconstant path $z(s)$ contains infinitely many distinct points accumulating inside the chart. Indeed one may restrict it to a closed interior interval on which it remains nonconstant. The inverse image $\Phi^{-1}(X)$ is a complex analytic subset of a connected one-dimensional chart. The identity theorem therefore makes it the entire chart. Continuing the defining homogeneous polynomial identities along S gives

$$\tilde{\Phi}(\tilde{S}) \subset X,$$

where $\tilde{S}$ is the connected universal cover and the flat trivialization defining $\tilde{\Phi}$ extends the one in the chosen chart.

Fix a period line L in this image. Equivariance places its entire monodromy orbit ΓL in X . By [Proposition 2.3](#), Γ is Zariski dense in $\mathrm{SO}(V_{\mathbb{C}}, b)$, and this group acts transitively on the nonsingular isotropic quadric Q_V . The orbit ΓL is therefore Zariski dense in Q_V . This contradicts (6.8) and $\dim Q_V = 3$.

Degrees in (6.5) are thus bounded. That inequality also bounds their ample heights, and Northcott's theorem on the fixed projective curve $\tilde{S}$ proves finiteness. $\square$

6.3 Descent to the coarse curve

Proof of Theorem 1.1. Suppose the stated locus on C were infinite. By Lemma 2.1, normalize C and take a connected component of a fixed fine-level cover containing infinitely many lifted target points. Removing the finitely many points needed for the smooth family and the algebraic charts leaves infinitely many. Choose an interior QM center and then the fixed auxiliary cover of Proposition 5.2. All these choices, including their fields of definition, are made once.

For every remaining target to which the construction is applied, Theorem 5.5 supplies a selected lift on this fixed cover with $h(t) \leq Ad(t)^\kappa$, for fixed A, κ . The degrees of a point and its lift are comparable by the fixed degree of the finite map, and the ample heights are comparable by the pullback of ample divisors, as in Lemma 2.1. Thus the constants are uniform as the quaternion algebra and the target vary.

Apply Proposition 6.2 on the fixed covering curve. Its counting argument uses the full orbit over the fixed field K : the height inequality for a selected lift is invariant under that orbit, and conjugation preserves its two integral Hodge vectors and their Rosati bounds. A conjugate marking need not satisfy the incidence imposed relative to the chosen center. That incidence was used to prove the height inequality; the incidence (6.7) depends only on the target's Hodge vectors. Hence all these selected lifts belong to the finite set of Proposition 6.2, whose final step uses Northcott's theorem. Their images on C form a finite set. Adding back the finitely many removed points contradicts the assumed infinitude. This proves the asserted finiteness for full geometric endomorphism algebra an indefinite quaternion division algebra over $\mathbb{Q}$, with no restriction on its discriminant or order. $\square$

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